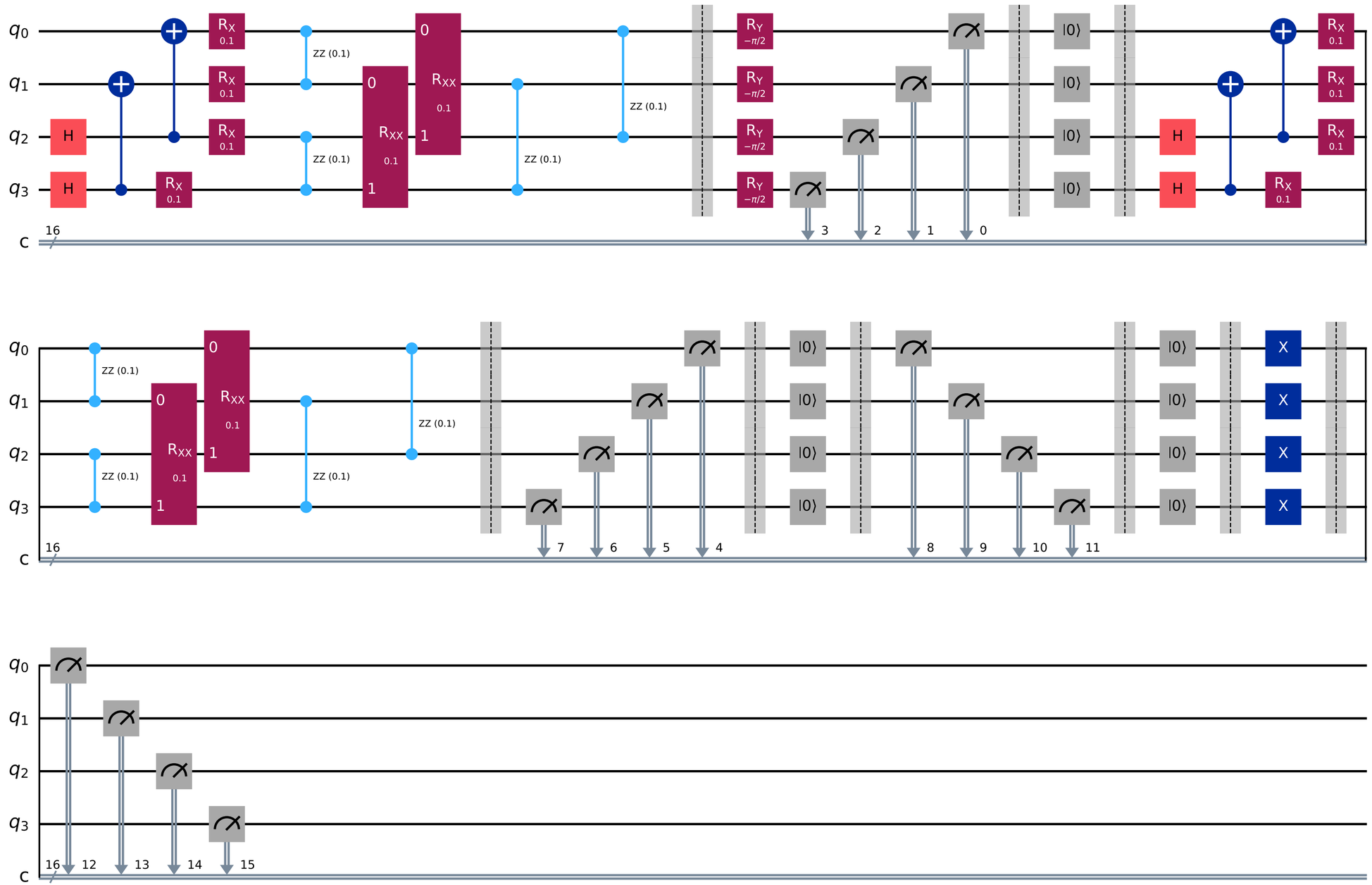
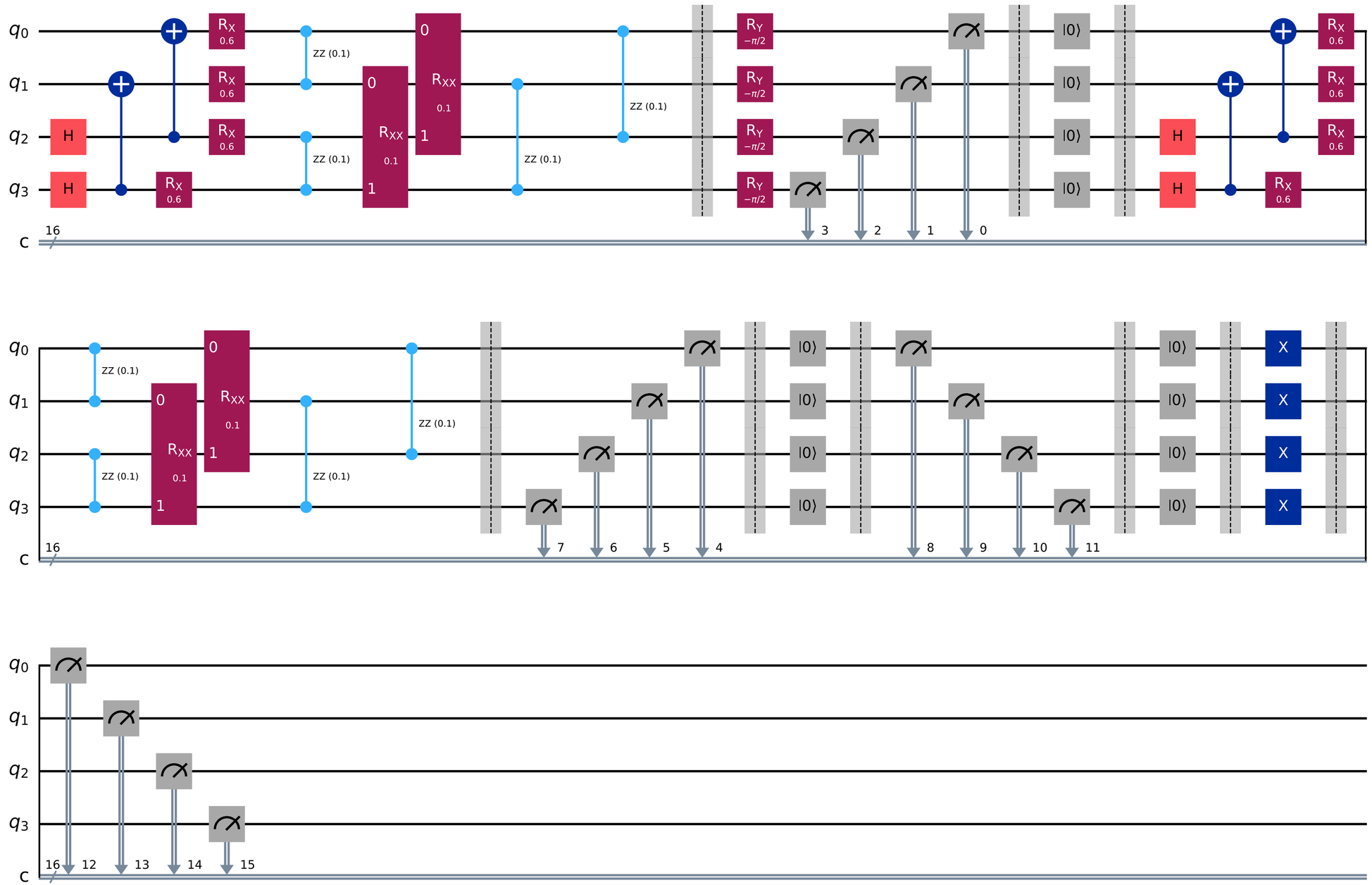
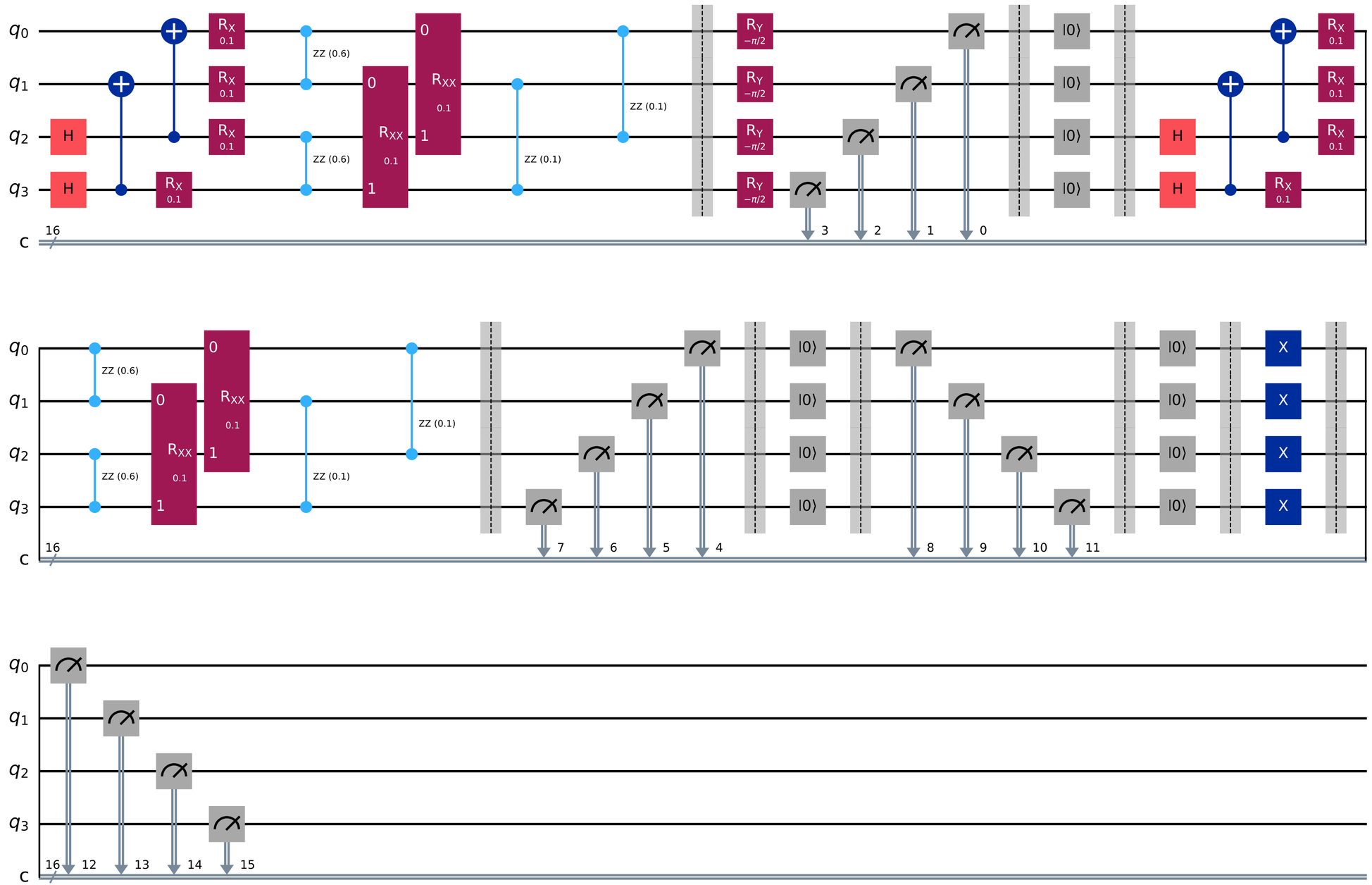
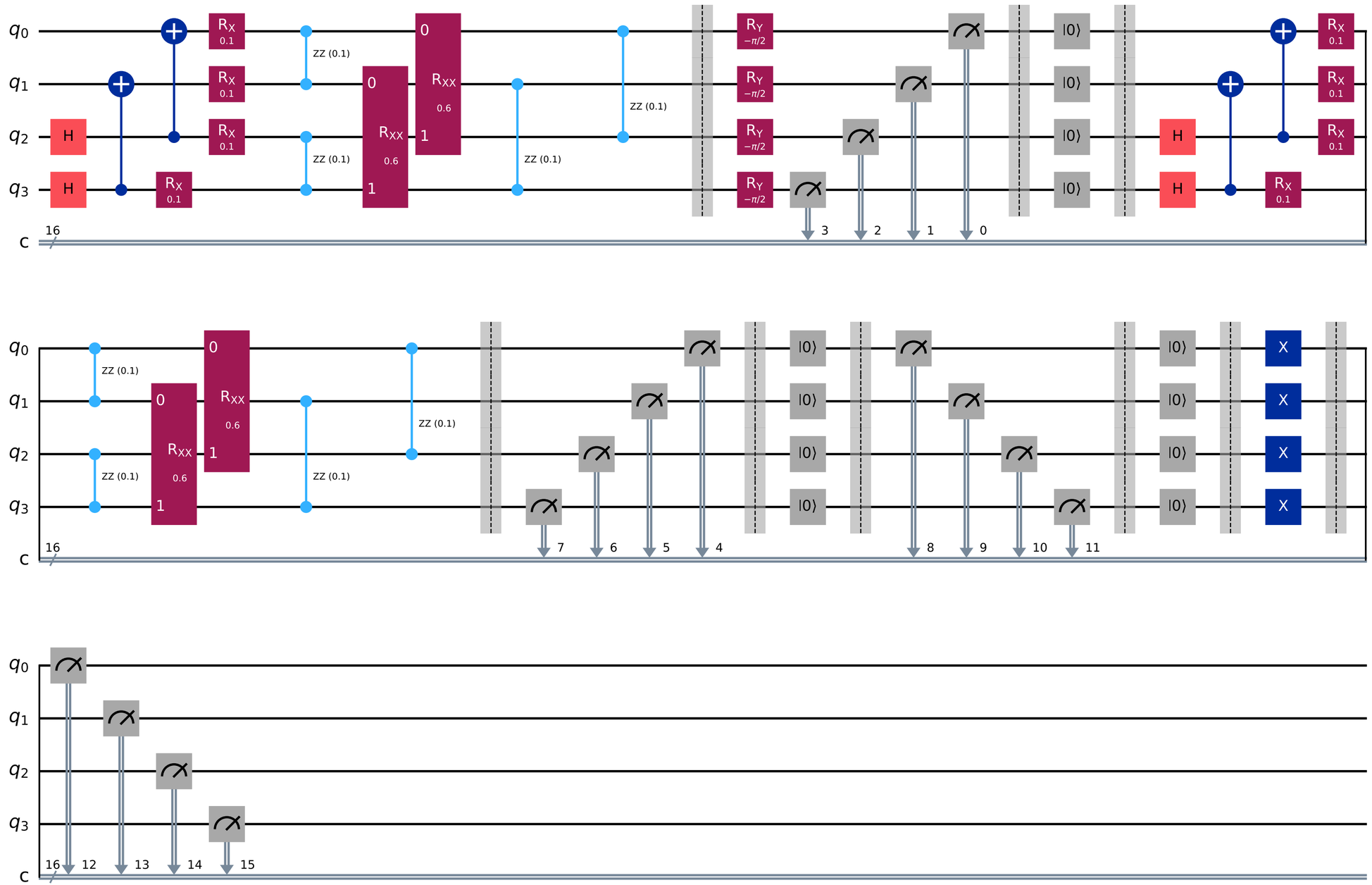
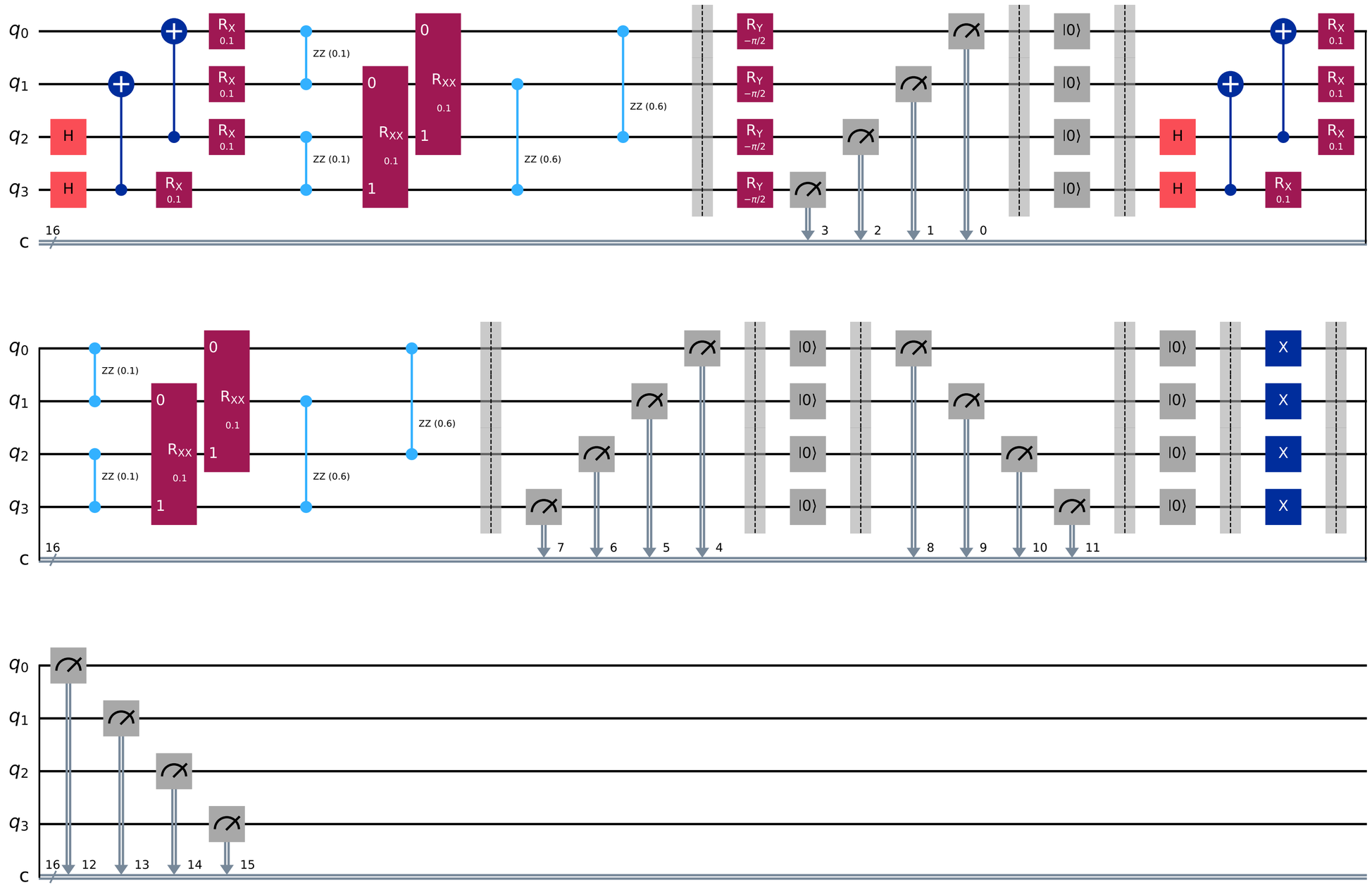
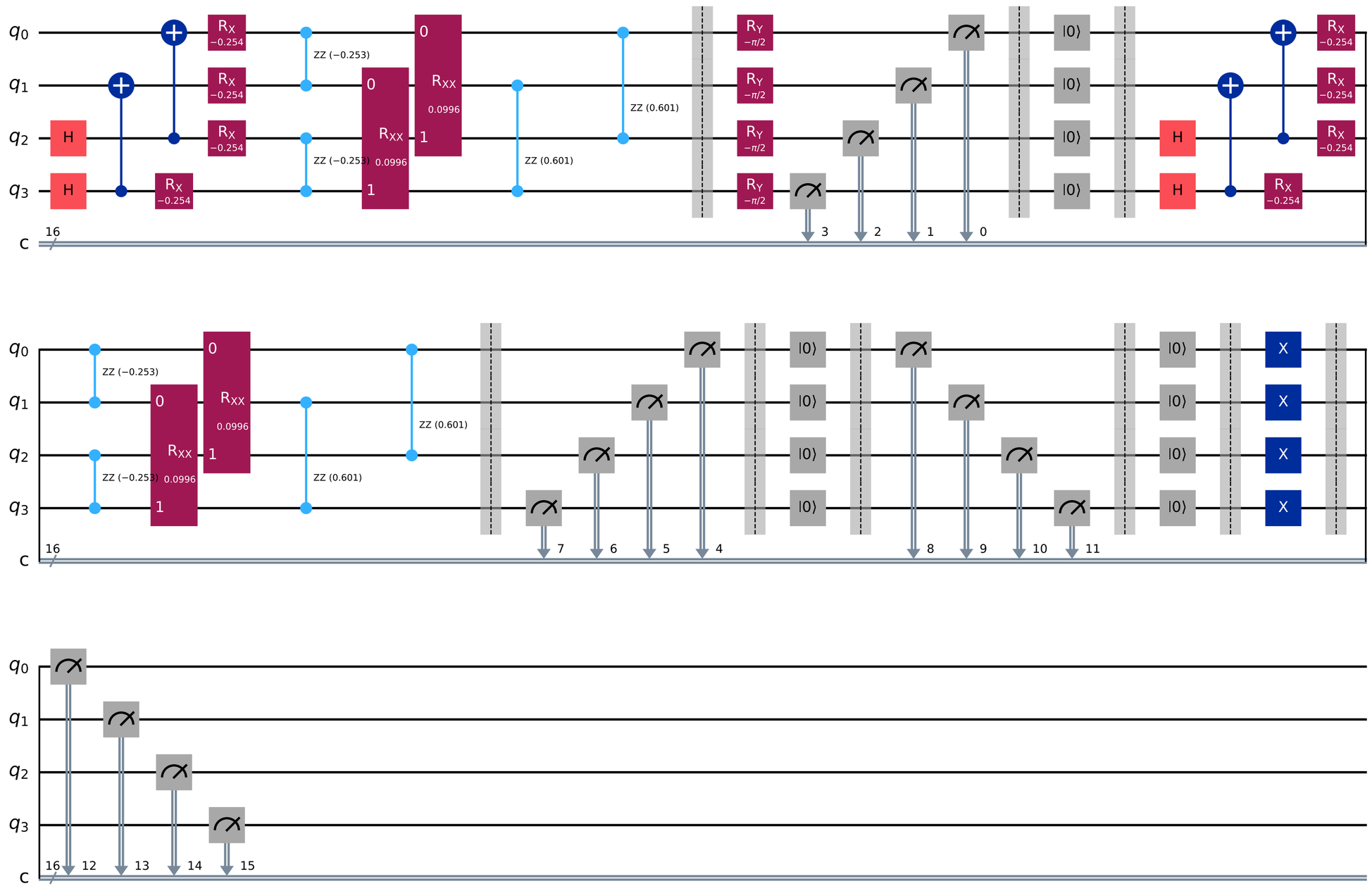

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

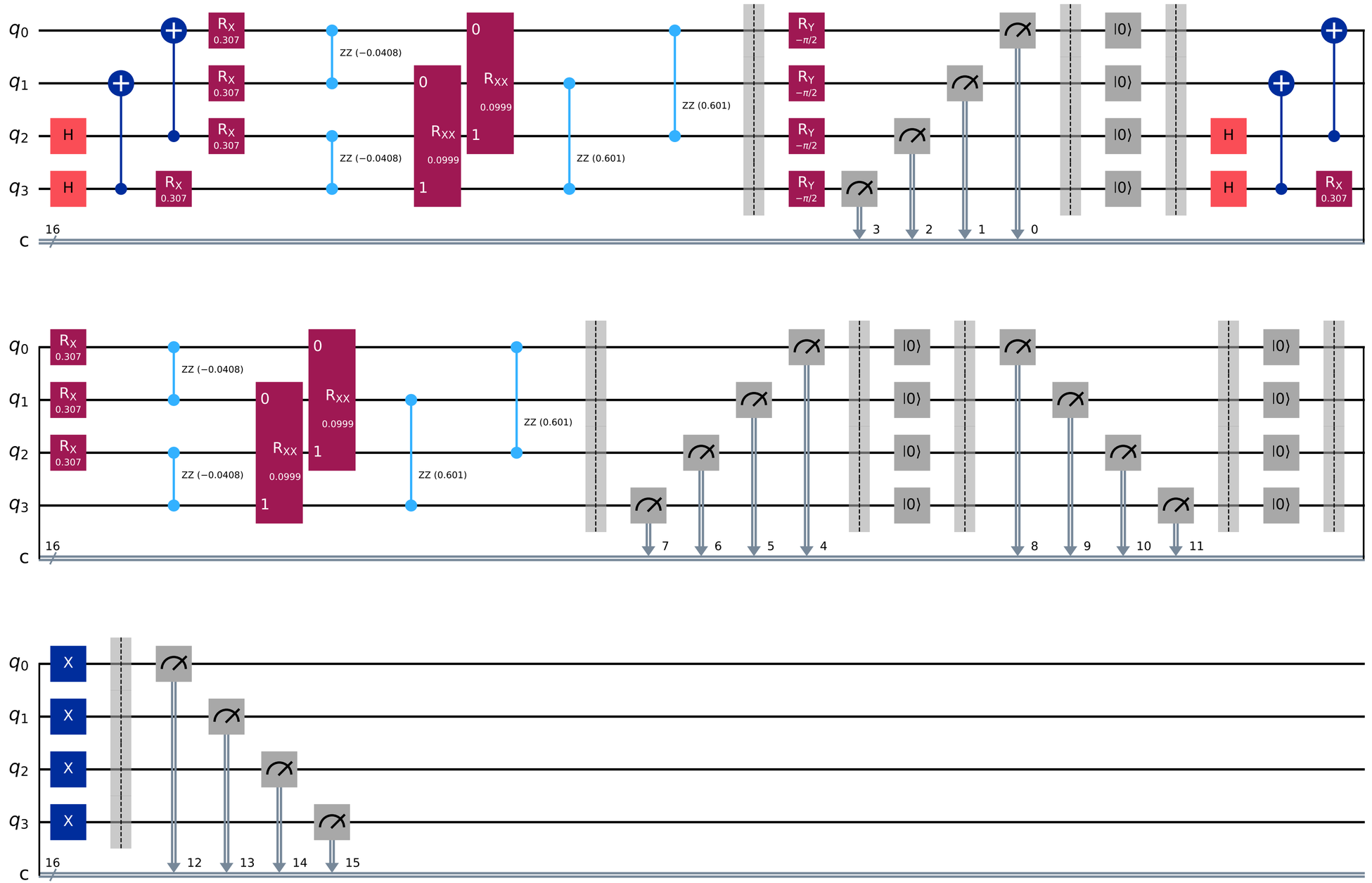

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.6
R_{XX} 0.1
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

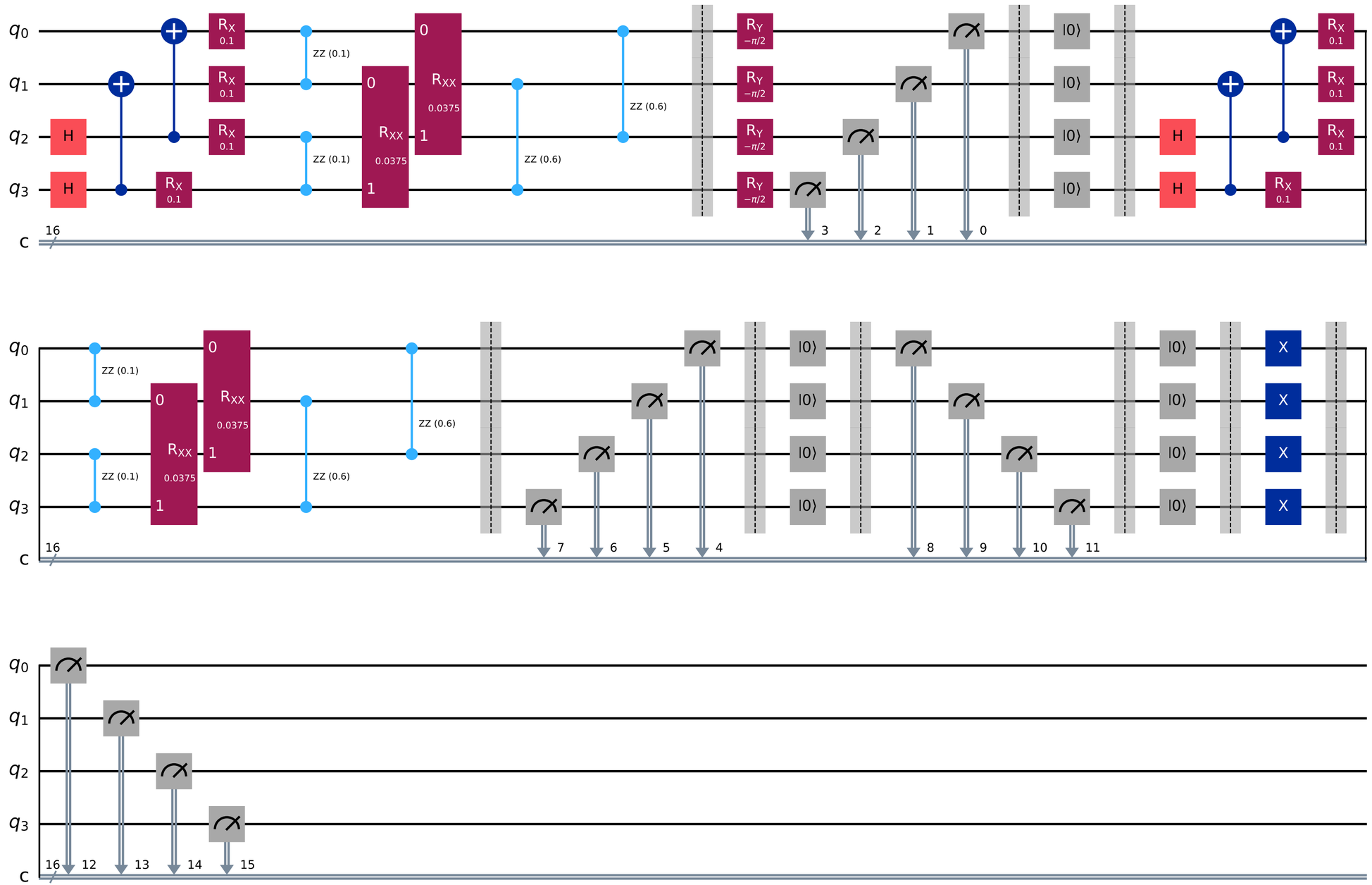

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.6)
ZZ (0.1)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

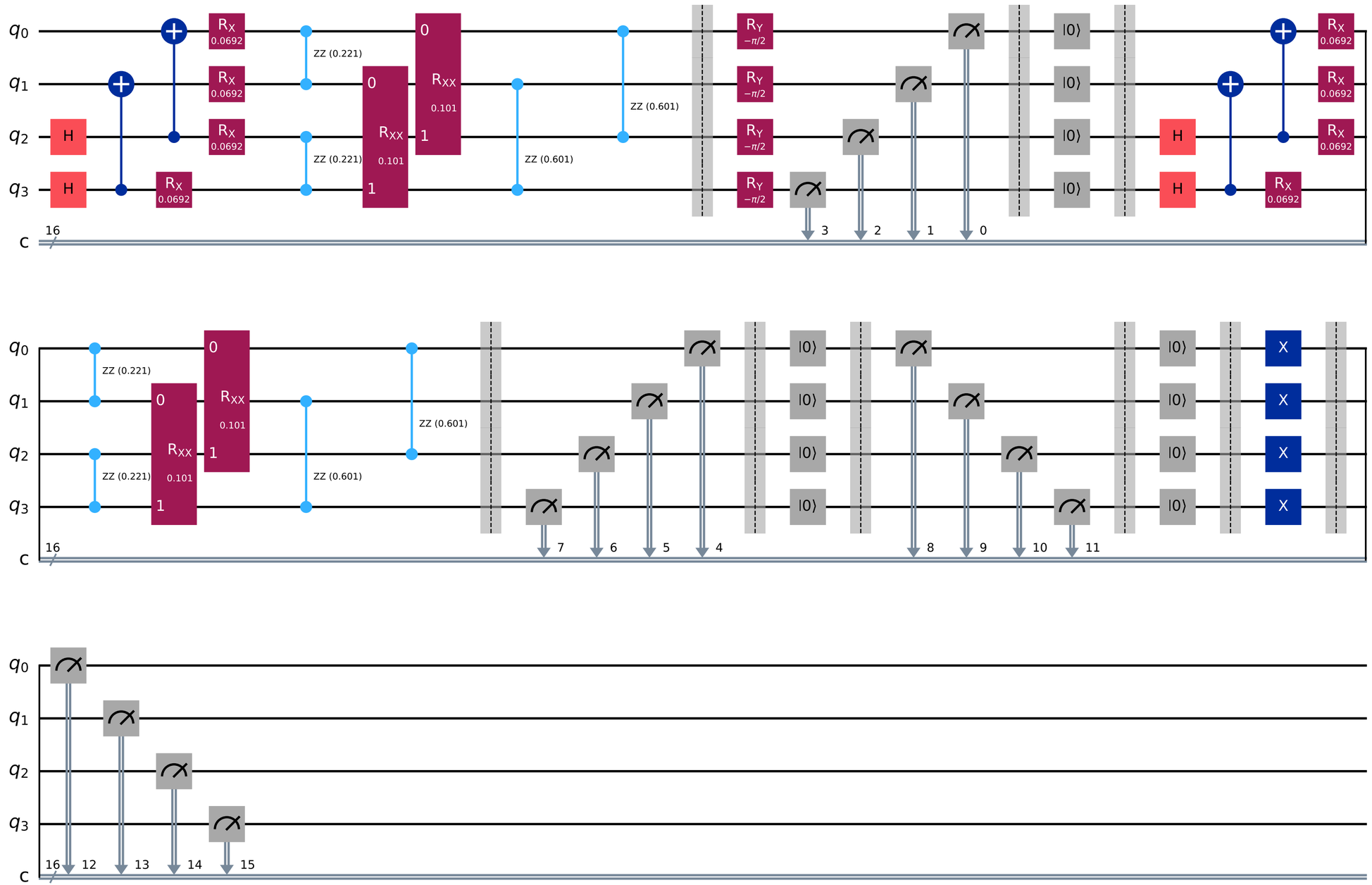

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_X 0.1
R_X 0.1
R_X 0.1
ZZ (0.1)
ZZ (0.1)
0
R_{XX} 0.6
1
0
R_{XX} 0.6
1
ZZ (0.1)
ZZ (0.1)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.1
R_X 0.1
R_X 0.1
R_X 0.1
ZZ (0.1)
ZZ (0.1)
0
R_{XX} 0.6
1
0
R_{XX} 0.6
1
ZZ (0.1)
ZZ (0.1)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

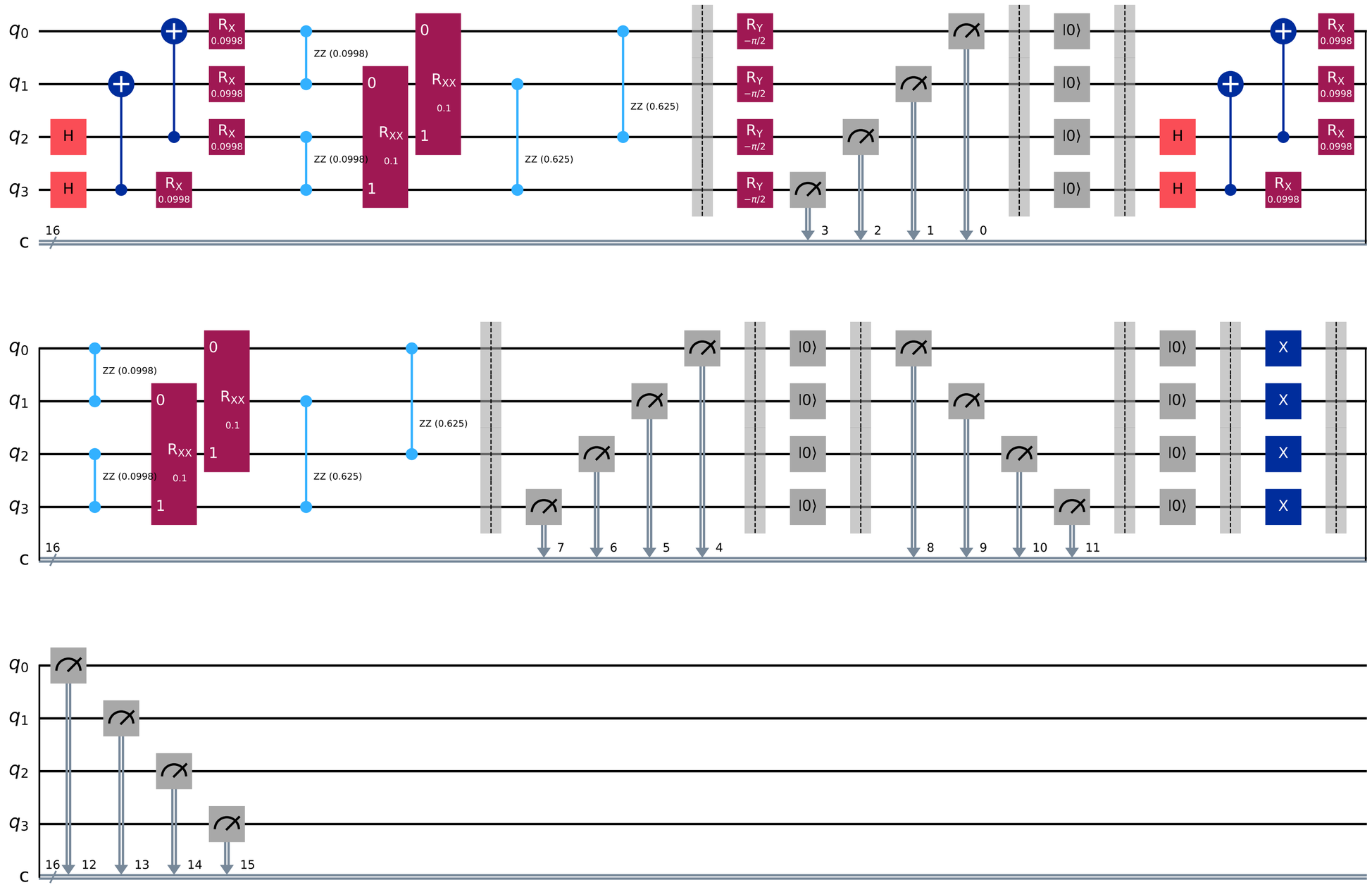

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_{XX} 0.1
ZZ (0.1)
ZZ (0.6)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

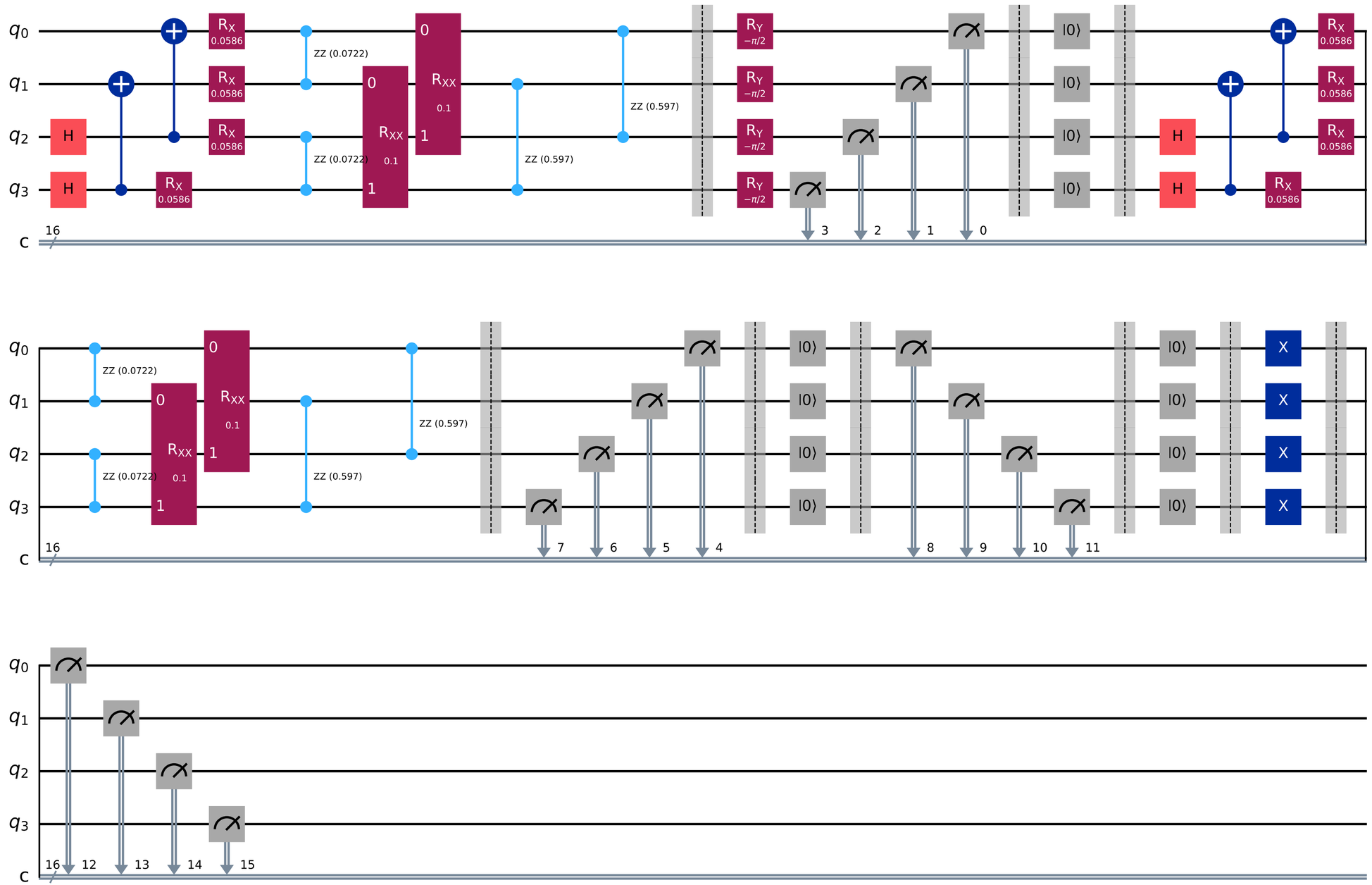

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.254
R_X −0.254
R_X −0.254
R_X −0.254
ZZ (−0.253)
ZZ (−0.253)
R_{XX} 0.0996
ZZ (0.601)
ZZ (0.601)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.254
R_X −0.254
R_X −0.254
R_X −0.254
ZZ (−0.253)
ZZ (−0.253)
R_{XX} 0.0996
ZZ (0.601)
ZZ (0.601)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

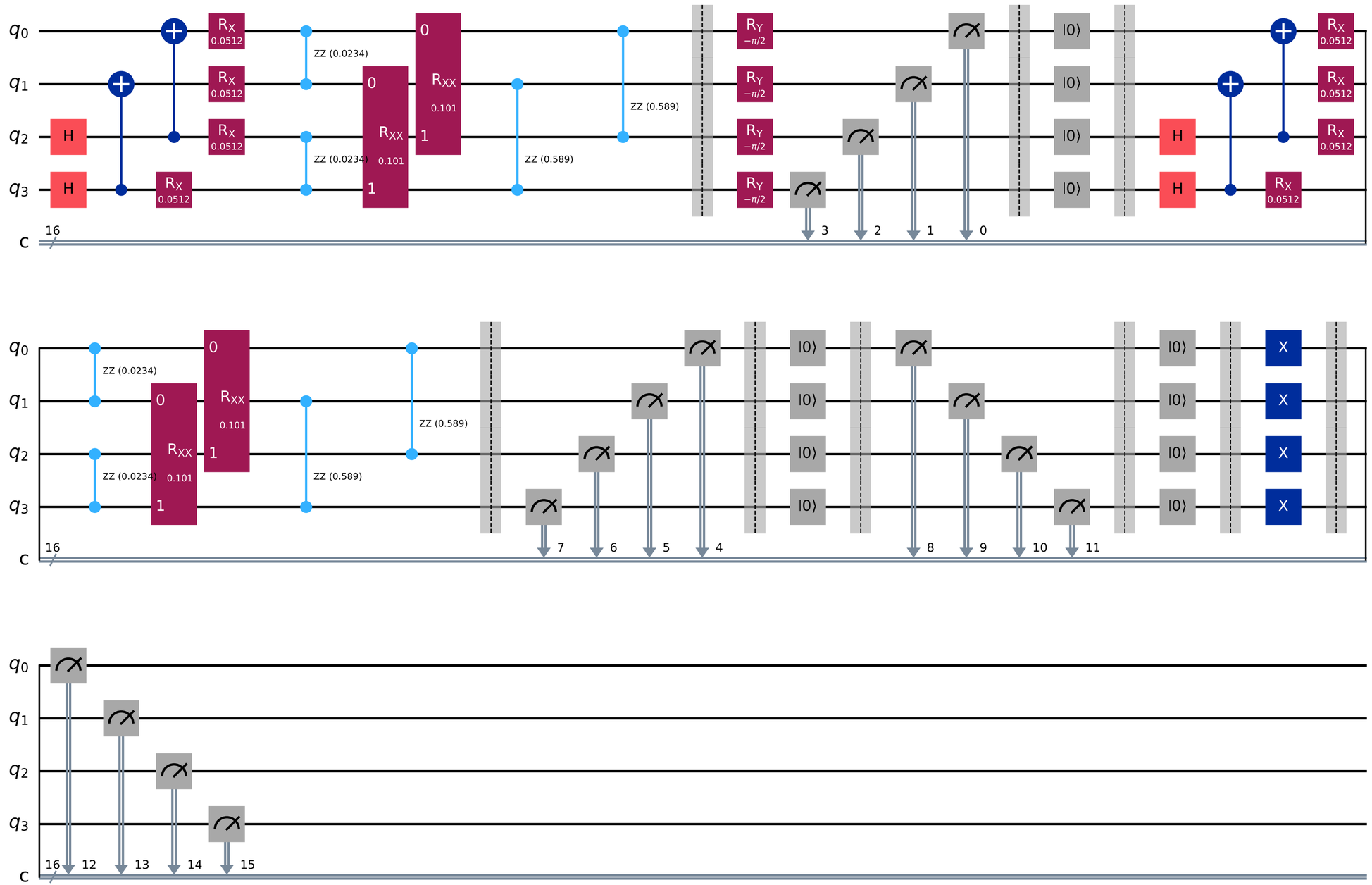

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.307
R_X 0.307
R_X 0.307
R_X 0.307
ZZ (−0.0408)
ZZ (−0.0408)
0
R_{XX} 0.0999
1
0
R_{XX} 0.0999
1
ZZ (0.601)
ZZ (0.601)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.307
R_X 0.307
R_X 0.307
R_X 0.307
ZZ (−0.0408)
ZZ (−0.0408)
0
R_{XX} 0.0999
1
0
R_{XX} 0.0999
1
ZZ (0.601)
ZZ (0.601)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

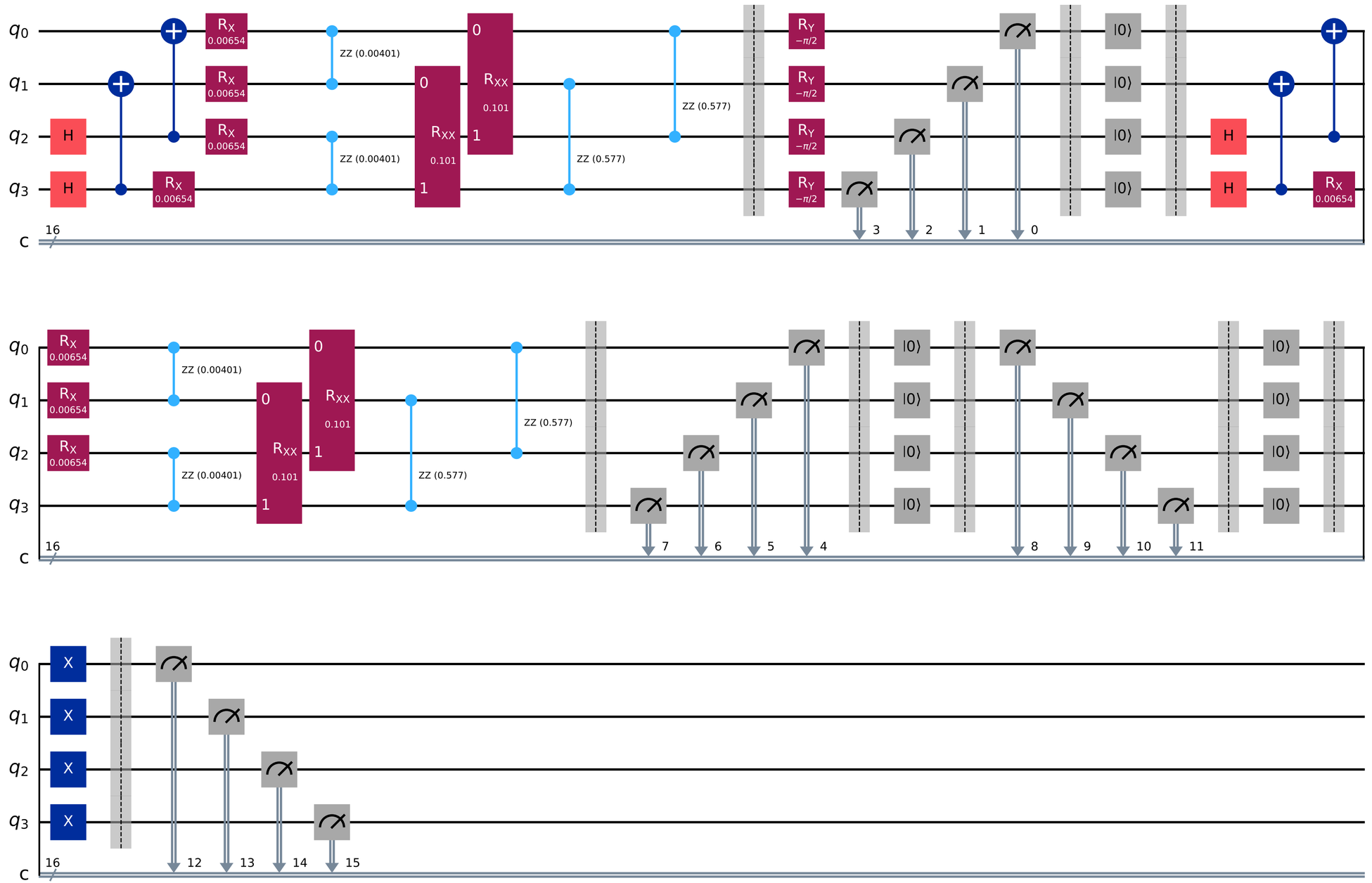

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.1
R_X 0.1
R_X 0.1
R_X 0.1
ZZ (0.1)
ZZ (0.1)
0
R_{XX} 0.0375
1
0
R_{XX} 0.0375
1
ZZ (0.6)
ZZ (0.6)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.1
R_X 0.1
R_X 0.1
R_X 0.1
ZZ (0.1)
ZZ (0.1)
0
R_{XX} 0.0375
1
0
R_{XX} 0.0375
1
ZZ (0.6)
ZZ (0.6)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

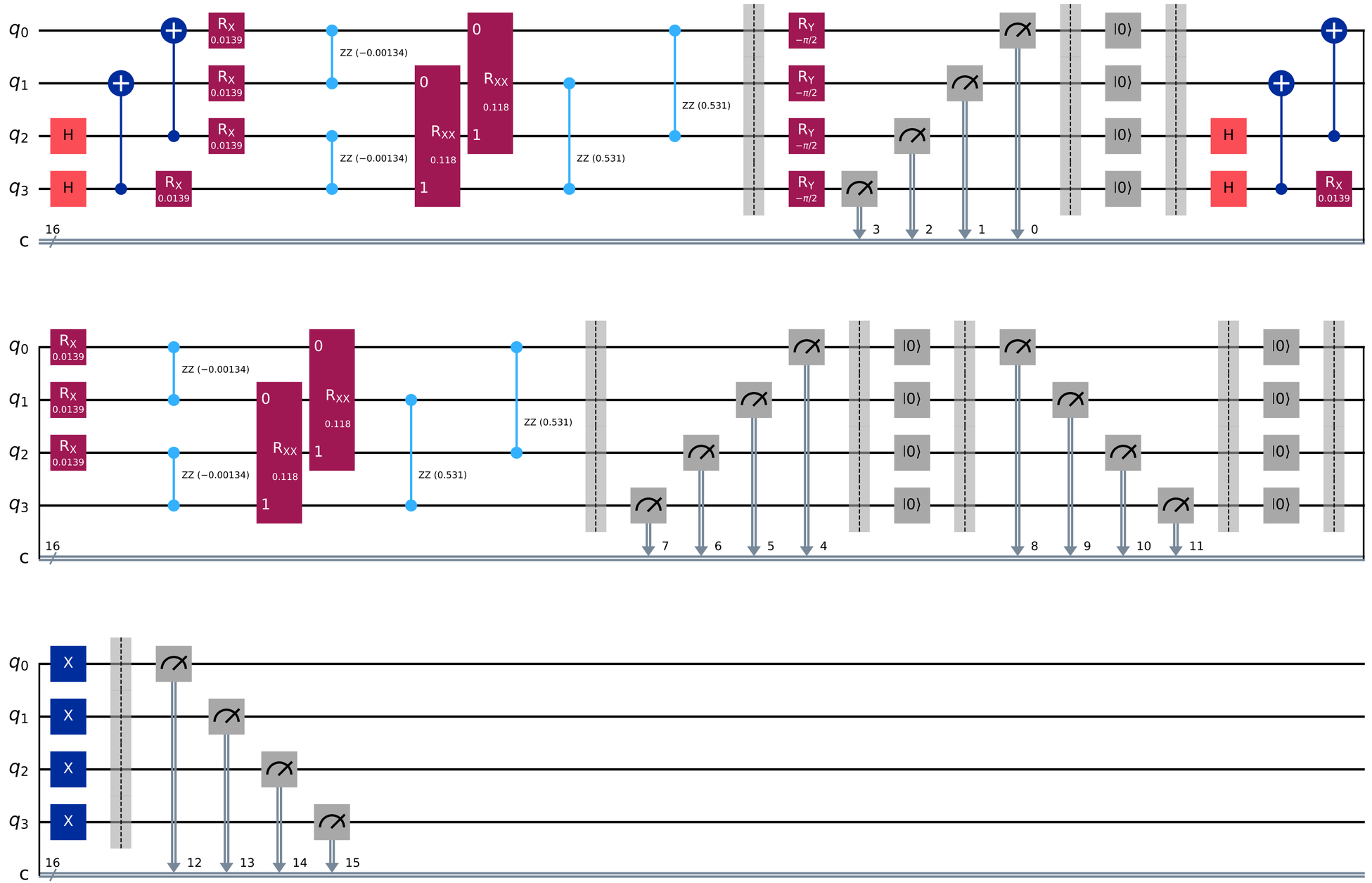

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0692
R_{XX} 0.101
ZZ (0.221)
ZZ (0.601)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

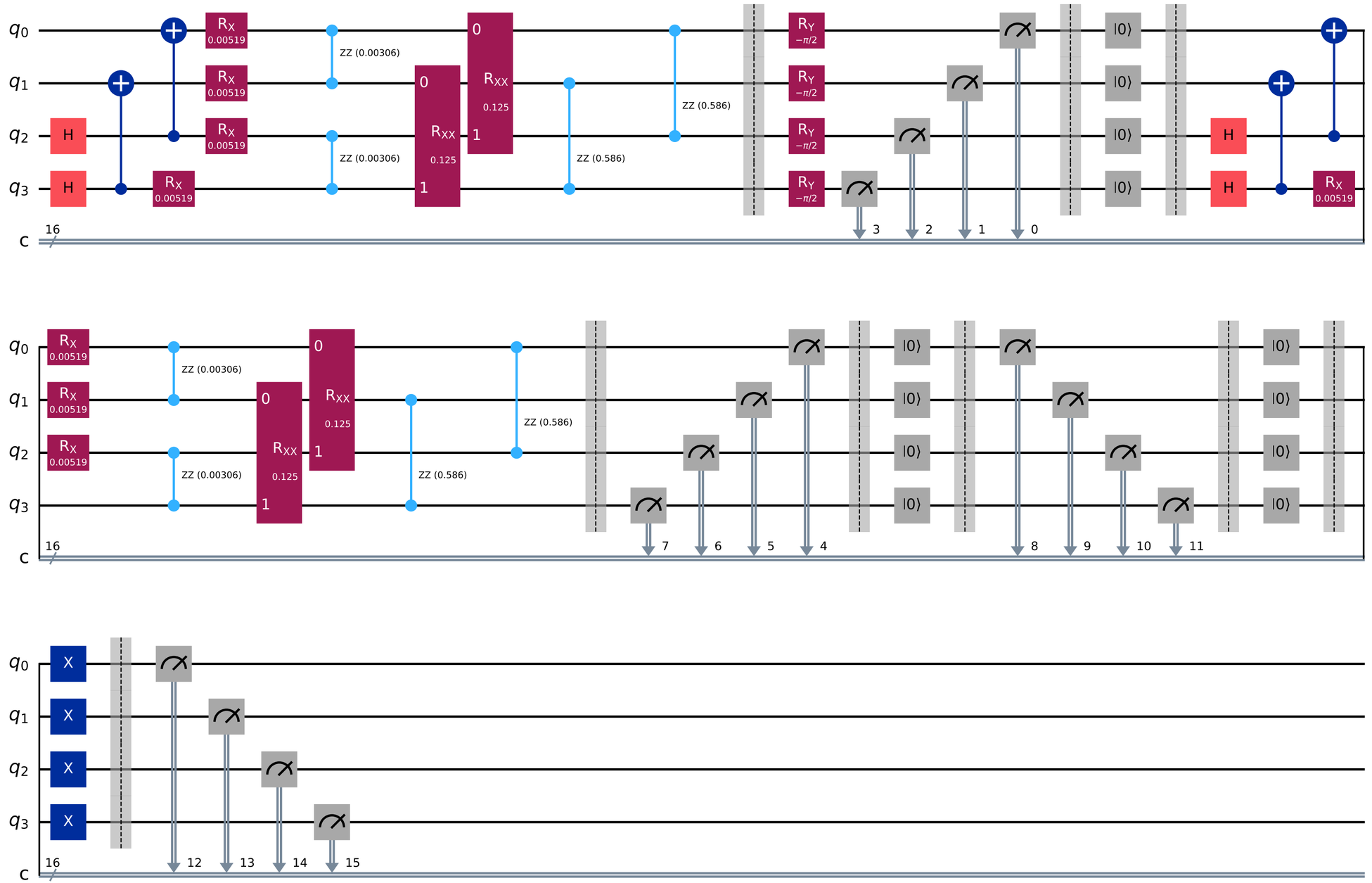

q0
q1
q2
q3
c
16
H
H
RX 0.0998
RX 0.0998
RX 0.0998
RX 0.0998
ZZ (0.0998)
ZZ (0.0998)
0
RXX 0.1
1
0
RXX 0.1
1
ZZ (0.625)
ZZ (0.625)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0998
RX 0.0998
RX 0.0998
RX 0.0998
ZZ (0.0998)
ZZ (0.0998)
0
RXX 0.1
1
0
RXX 0.1
1
ZZ (0.625)
ZZ (0.625)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

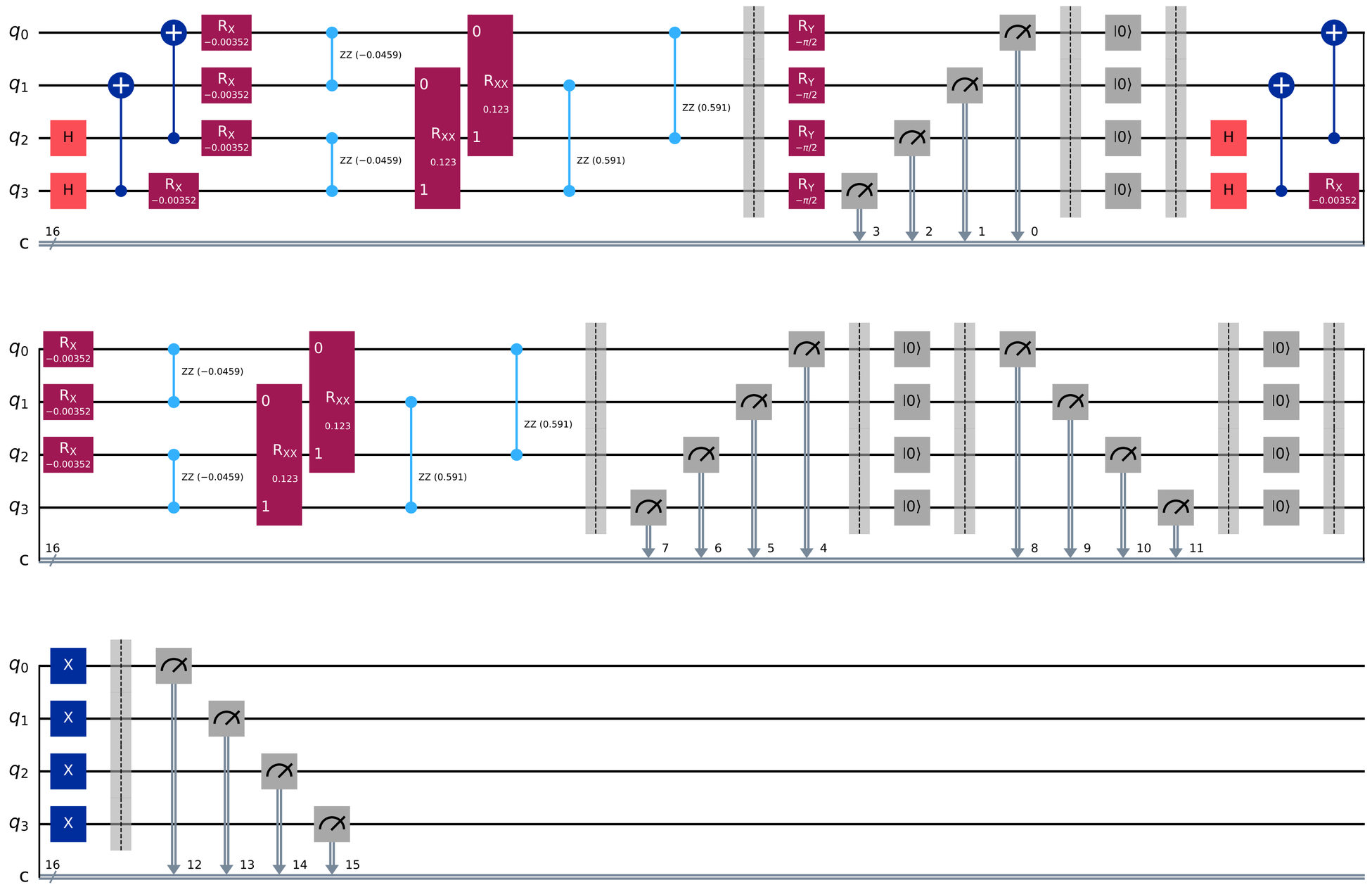

q0
q1
q2
q3
c
16
H
H
RX 0.0586
RX 0.0586
RX 0.0586
RX 0.0586
ZZ (0.0722)
ZZ (0.0722)
0
RXX 0.1
1
0
RXX 0.1
1
ZZ (0.597)
ZZ (0.597)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0586
RX 0.0586
RX 0.0586
RX 0.0586
ZZ (0.0722)
ZZ (0.0722)
0
RXX 0.1
1
0
RXX 0.1
1
ZZ (0.597)
ZZ (0.597)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
q0
q1
q2
q3
c
16
12
13
14
15

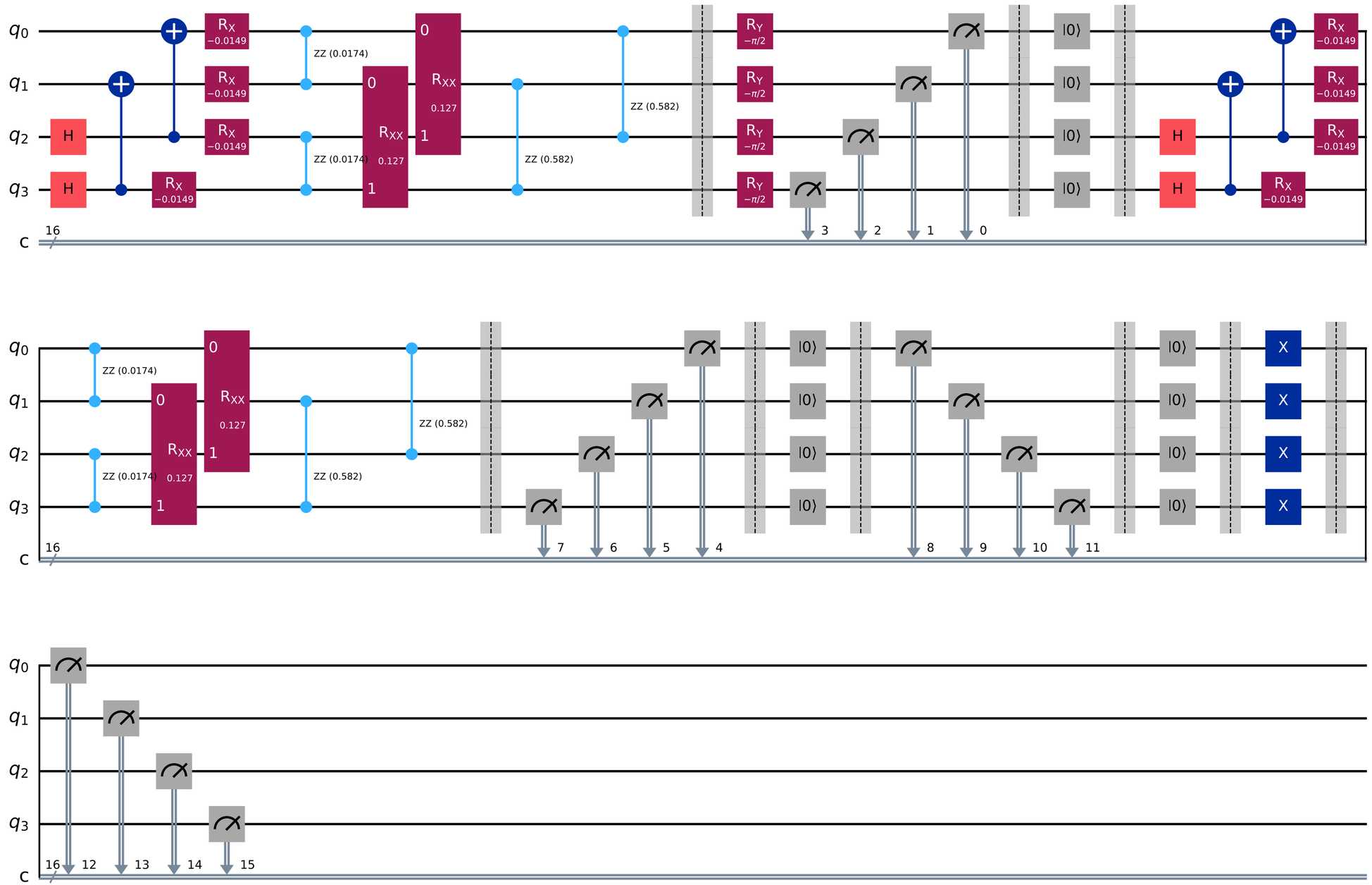

q0
q1
q2
q3
c
16
H
H
RX 0.0512
RX 0.0512
RX 0.0512
RX 0.0512
ZZ (0.0234)
ZZ (0.0234)
0
RXX 0.101
1
0
RXX 0.101
1
ZZ (0.589)
ZZ (0.589)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0512
RX 0.0512
RX 0.0512
RX 0.0512
ZZ (0.0234)
ZZ (0.0234)
0
RXX 0.101
1
0
RXX 0.101
1
ZZ (0.589)
ZZ (0.589)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

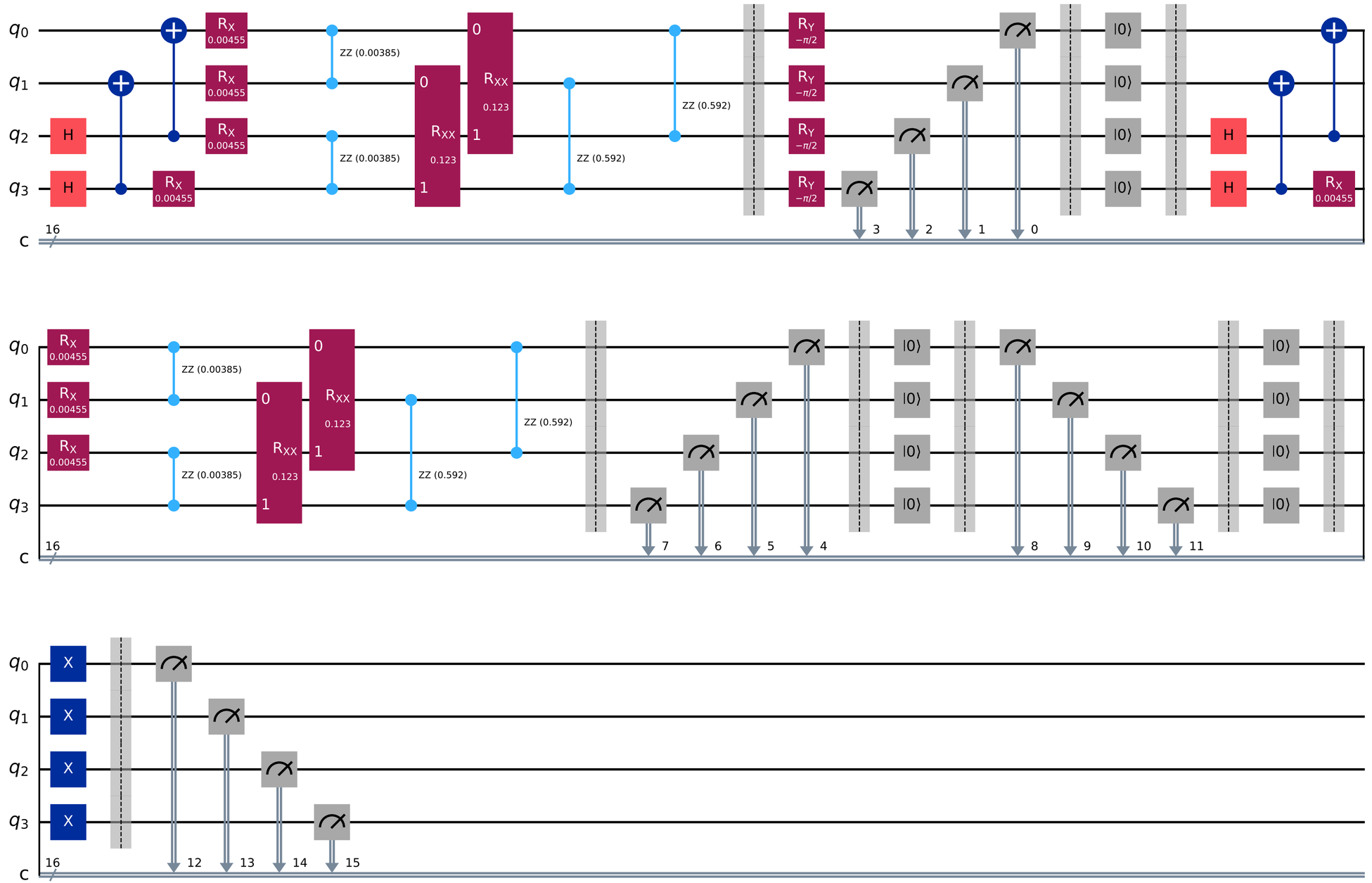

q0
q1
q2
q3
c
16
H
H
RX 0.00654
RX 0.00654
RX 0.00654
RX 0.00654
ZZ (0.00401)
ZZ (0.00401)
0
RXX 0.101
1
0
RXX 0.101
1
ZZ (0.577)
ZZ (0.577)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.00654
3
2
1
0
RX 0.00654
RX 0.00654
RX 0.00654
ZZ (0.00401)
ZZ (0.00401)
0
RXX 0.101
1
0
RXX 0.101
1
ZZ (0.577)
ZZ (0.577)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

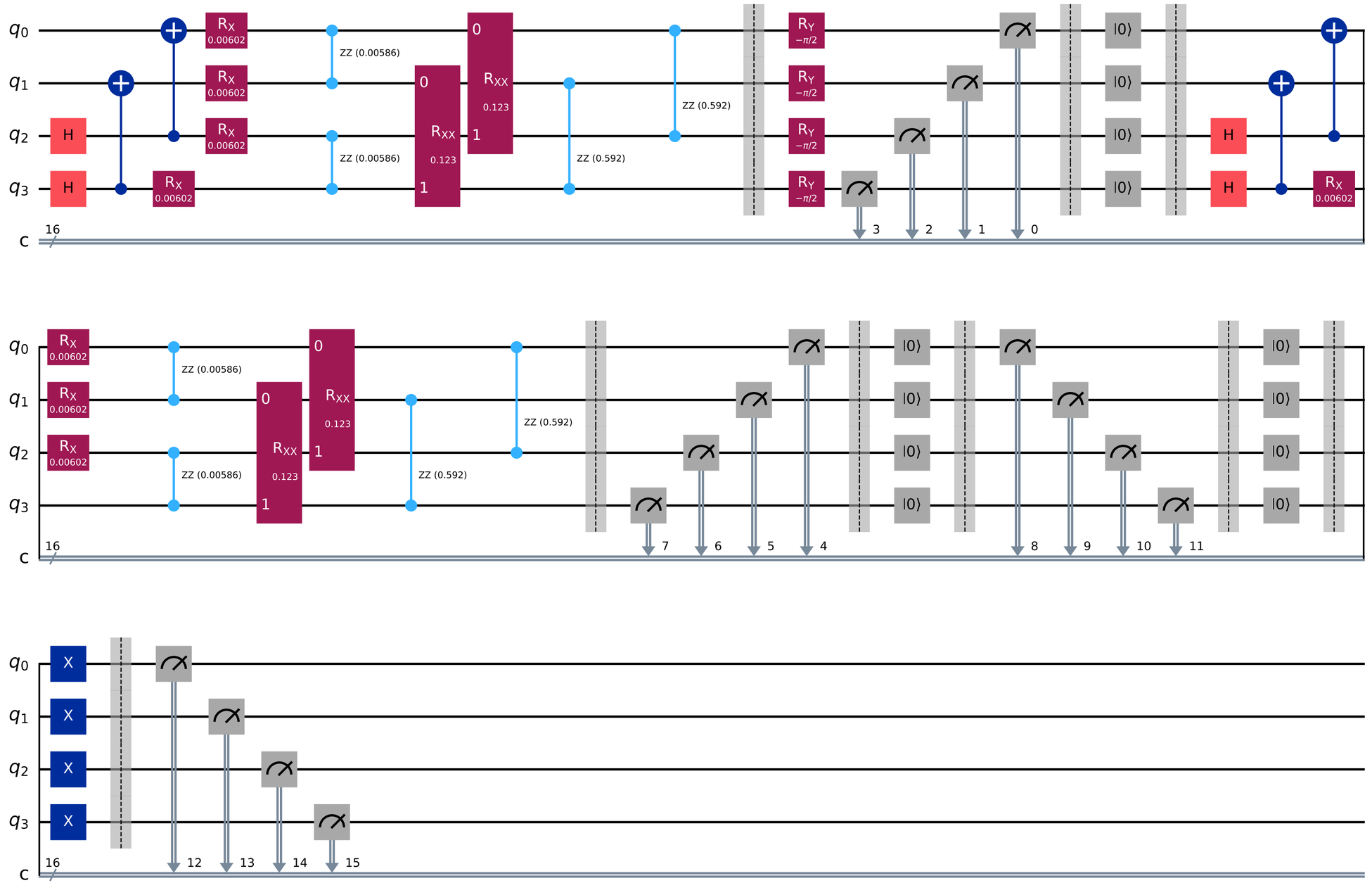

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0139
ZZ (−0.00134)
ZZ (0.531)
R_{XX} 0.118
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

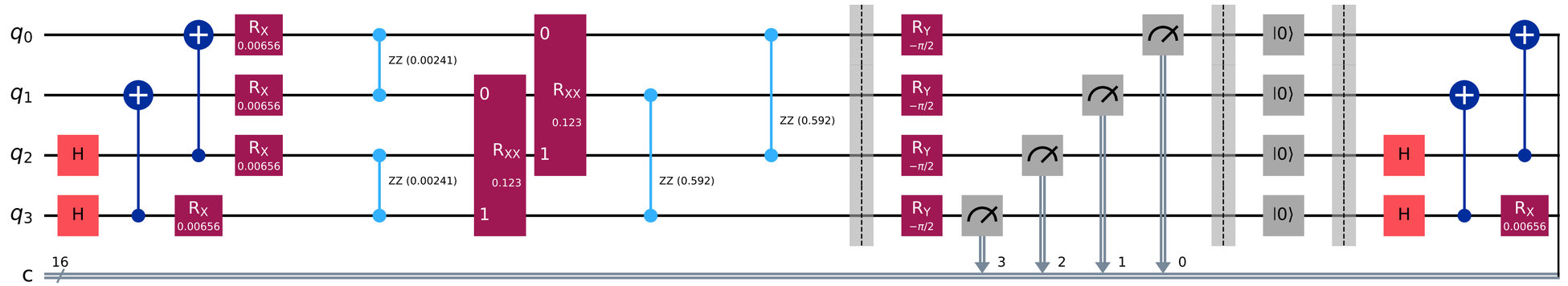

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00519
R_X 0.00519
R_X 0.00519
R_X 0.00519
ZZ (0.00306)
ZZ (0.00306)
0
R_{XX} 0.125
1
0
R_{XX} 0.125
1
ZZ (0.586)
ZZ (0.586)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.00519
3
2
1
0
R_X 0.00519
R_X 0.00519
R_X 0.00519
ZZ (0.00306)
ZZ (0.00306)
0
R_{XX} 0.125
1
0
R_{XX} 0.125
1
ZZ (0.586)
ZZ (0.586)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

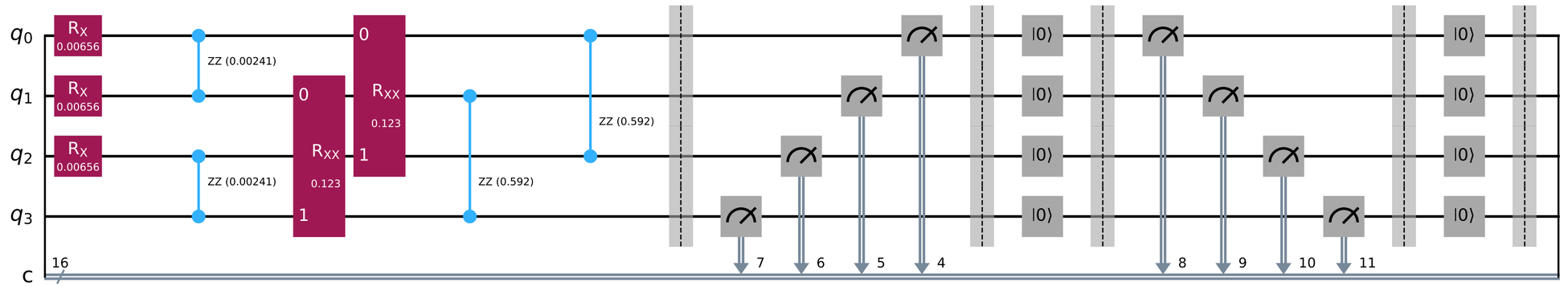

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.00352
R_X −0.00352
R_X −0.00352
R_X −0.00352
ZZ (−0.0459)
ZZ (−0.0459)
0
R_{XX}
0.123
1
0
R_{XX}
0.123
1
ZZ (0.591)
ZZ (0.591)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.00352
3
2
1
0
R_X −0.00352
R_X −0.00352
R_X −0.00352
ZZ (−0.0459)
ZZ (−0.0459)
0
R_{XX}
0.123
1
ZZ (0.591)
ZZ (0.591)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

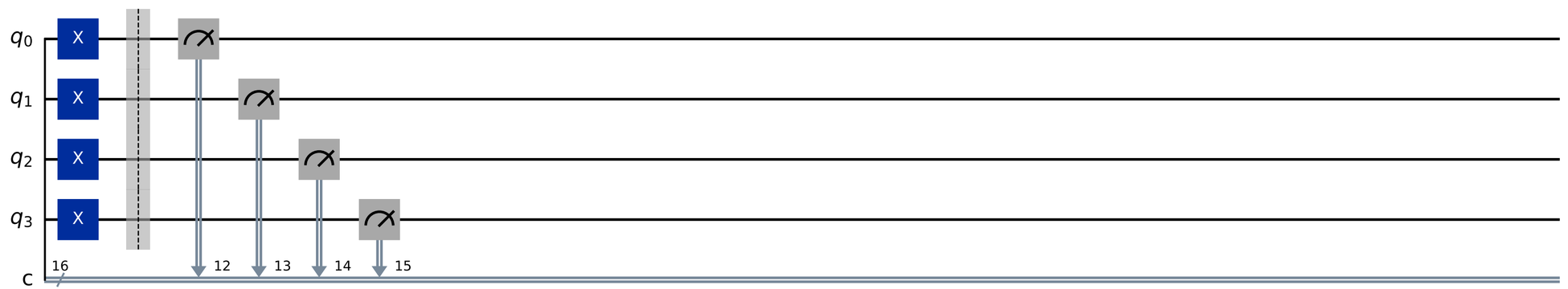

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0149
R_X −0.0149
R_X −0.0149
R_X −0.0149
ZZ (0.0174)
ZZ (0.0174)
0
R_{XX}
0.127
1
0
R_{XX}
0.127
1
ZZ (0.582)
ZZ (0.582)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0149
R_X −0.0149
R_X −0.0149
R_X −0.0149
ZZ (0.0174)
ZZ (0.0174)
0
R_{XX}
0.127
1
0
R_{XX}
0.127
1
ZZ (0.582)
ZZ (0.582)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

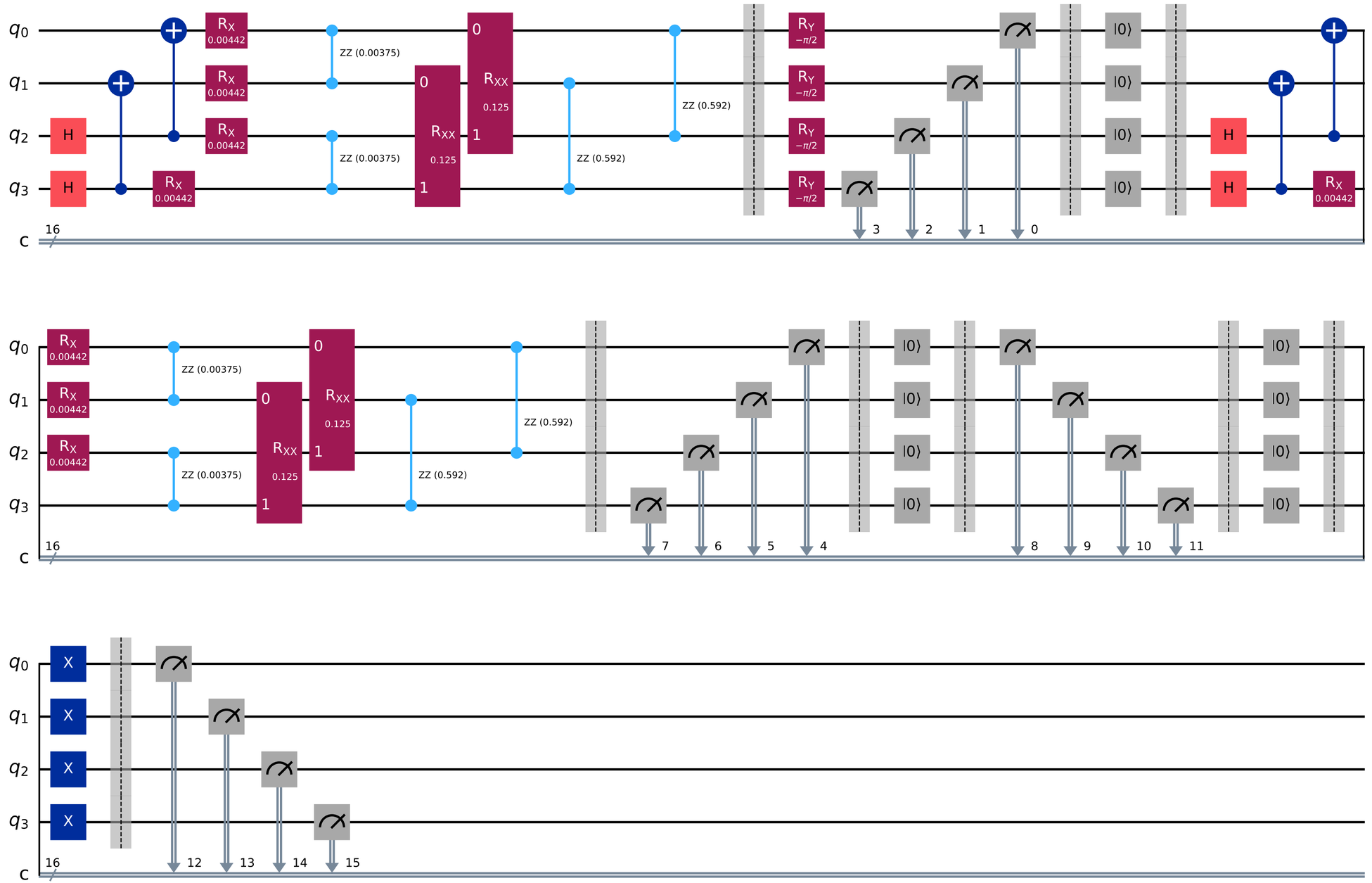

q0
q1
q2
q3
c
16
H
H
RX
0.00455
ZZ (0.00385)
ZZ (0.00385)
RXX
0.123
ZZ (0.592)
ZZ (0.592)
RY
−π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X

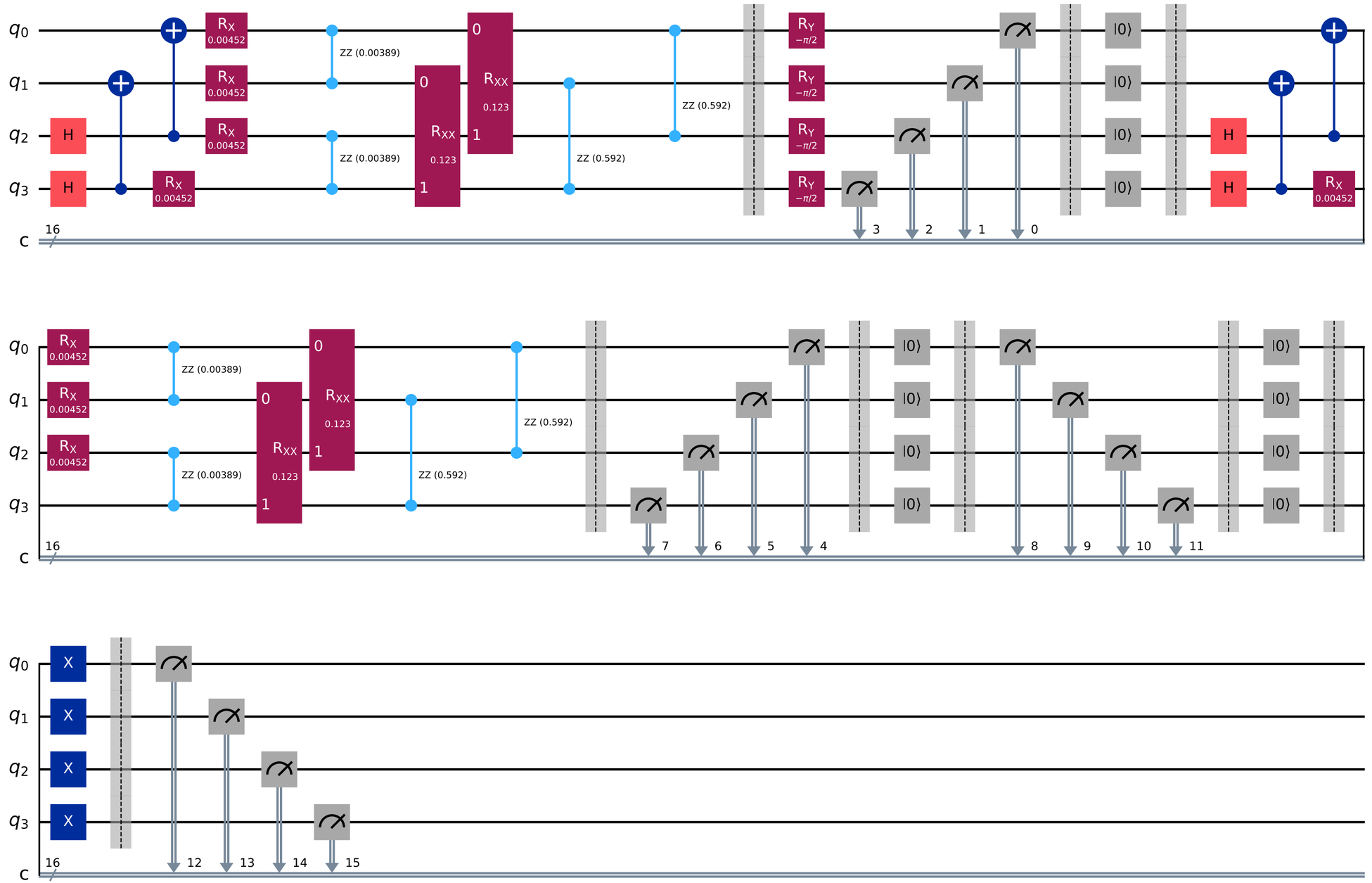

q0
q1
q2
q3
c
16
H
H
RX
0.00602
ZZ (0.00586)
ZZ (0.592)
RXX
0.123
RY
−π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X

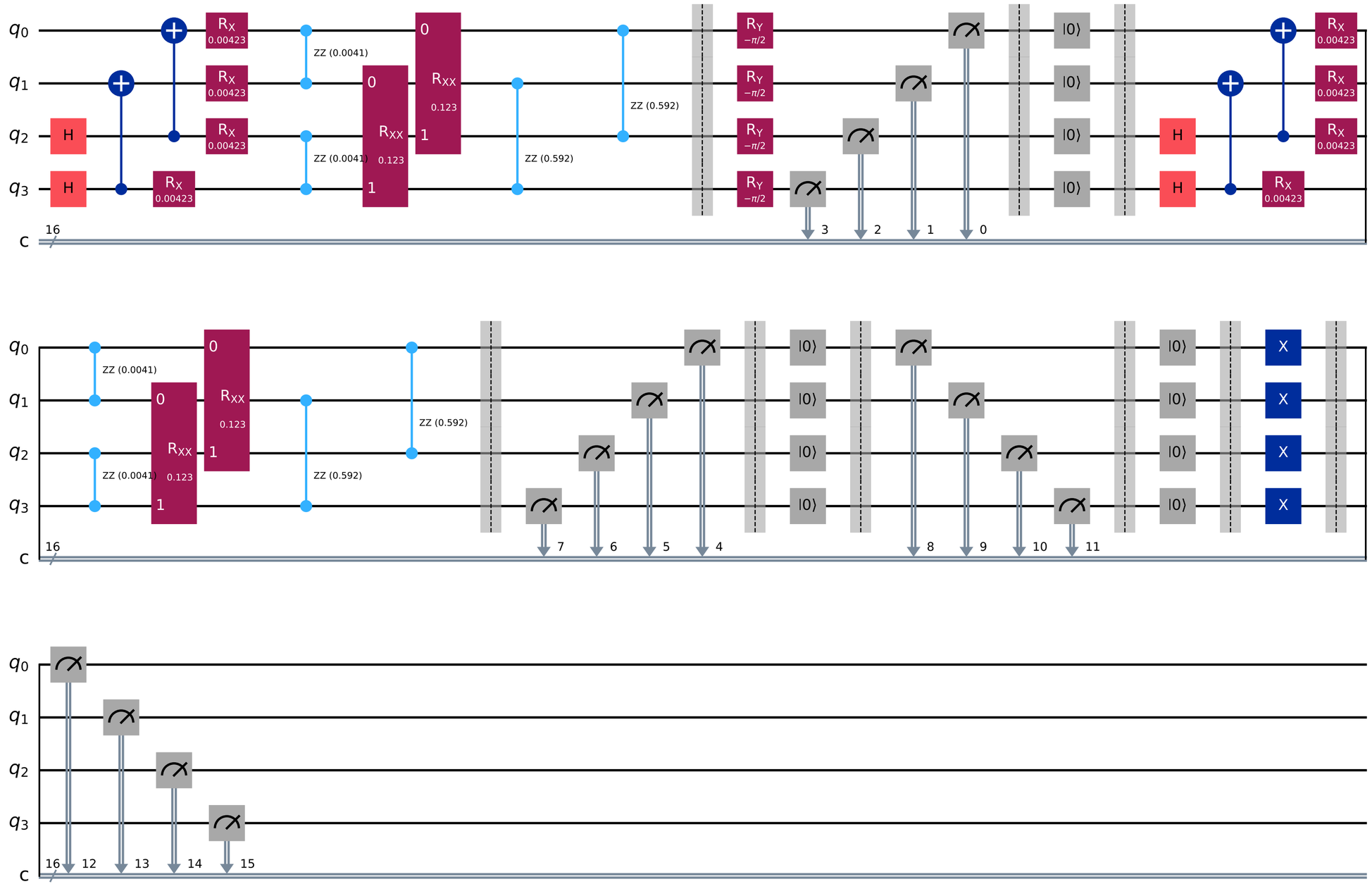

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00656
R_X 0.00656
R_X 0.00656
R_X 0.00656
ZZ (0.00241)
ZZ (0.00241)
R_{XX} 0.123
ZZ (0.592)
ZZ (0.592)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.00656
3
2
1
0

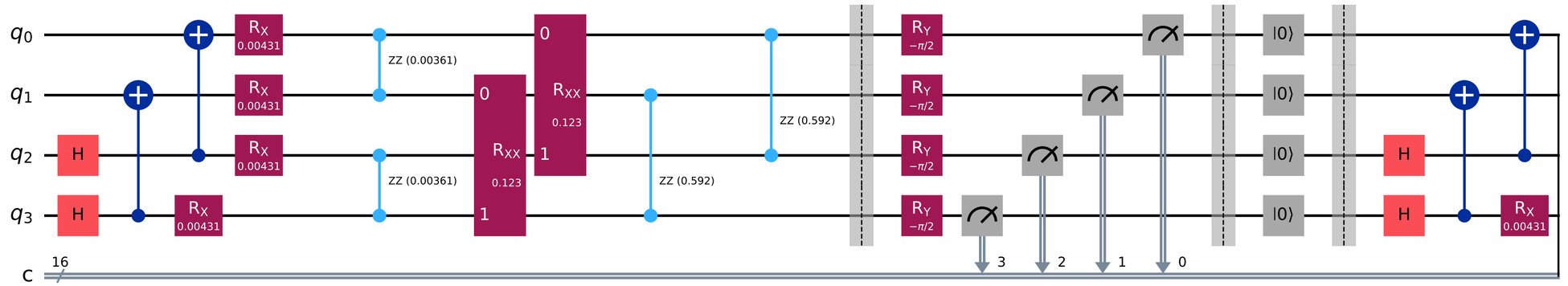

q_0
q_1
q_2
q_3
c
16
R_X 0.00656
R_X 0.00656
R_X 0.00656
ZZ (0.00241)
ZZ (0.00241)
R_{XX} 0.123
R_{XX} 0.123
ZZ (0.592)
ZZ (0.592)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11

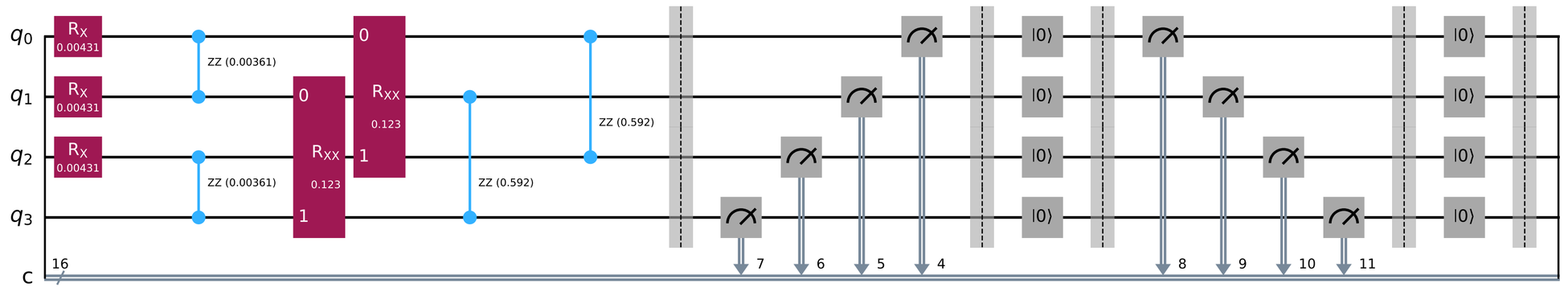

q_0
q_1
q_2
q_3
c
16
X
X
X
X
12
13
14
15

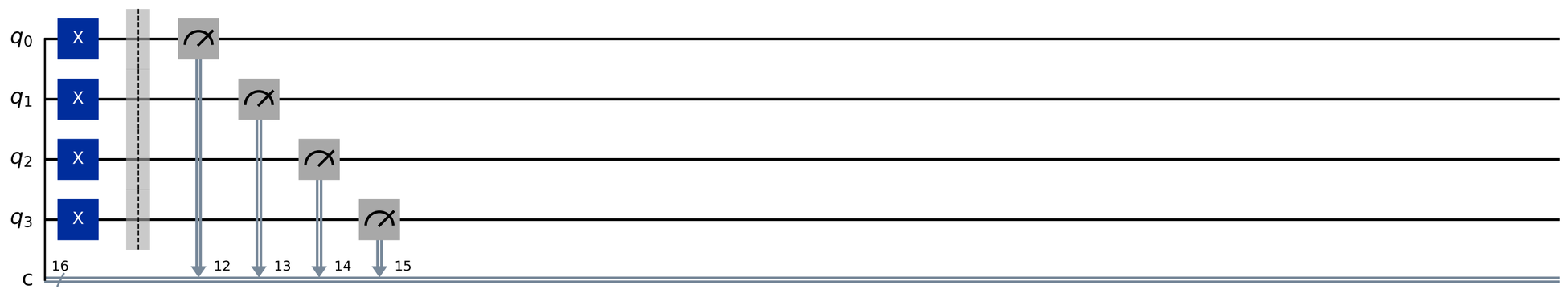

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00442
R_X 0.00442
R_X 0.00442
R_X 0.00442
ZZ (0.00375)
ZZ (0.00375)
R_{XX} 0.125
R_{XX} 0.125
ZZ (0.592)
ZZ (0.592)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.00442
3
2
1
0
R_X 0.00442
R_X 0.00442
R_X 0.00442
ZZ (0.00375)
ZZ (0.00375)
R_{XX} 0.125
R_{XX} 0.125
ZZ (0.592)
ZZ (0.592)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

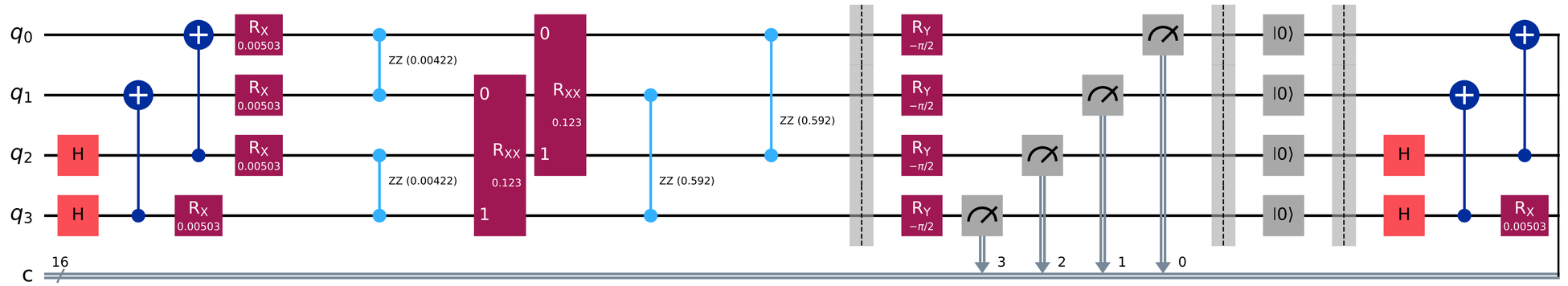

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00452
R_X 0.00452
R_X 0.00452
R_X 0.00452
ZZ (0.00389)
ZZ (0.00389)
R_{XX} 0.123
ZZ (0.592)
ZZ (0.592)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.00452
3
2
1
0
R_X 0.00452
R_X 0.00452
R_X 0.00452
ZZ (0.00389)
ZZ (0.00389)
R_{XX} 0.123
ZZ (0.592)
ZZ (0.592)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

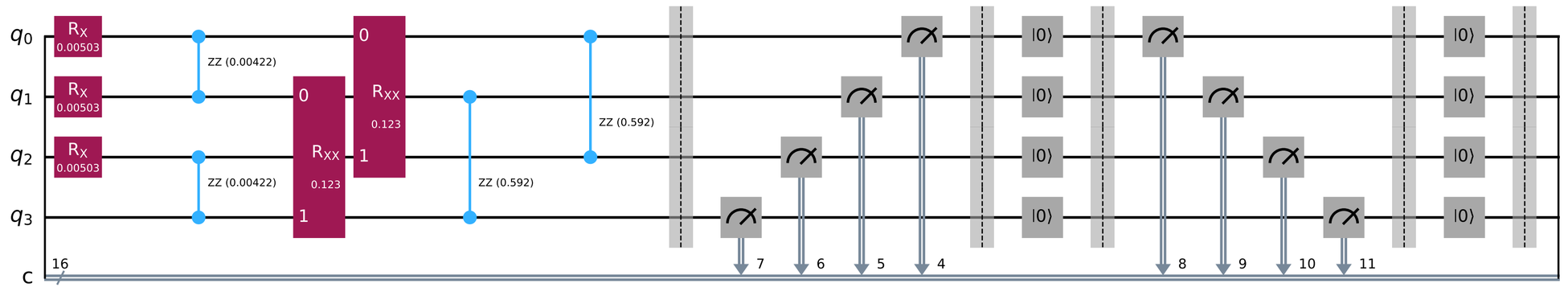

q0
q1
q2
q3
c
16
H
H
RX 0.00423
RX 0.00423
RX 0.00423
RX 0.00423
ZZ (0.0041)
ZZ (0.0041)
0
RXX 0.123
1
ZZ (0.592)
ZZ (0.592)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.00423
RX 0.00423
RX 0.00423
RX 0.00423
ZZ (0.0041)
ZZ (0.0041)
0
RXX 0.123
1
ZZ (0.592)
ZZ (0.592)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

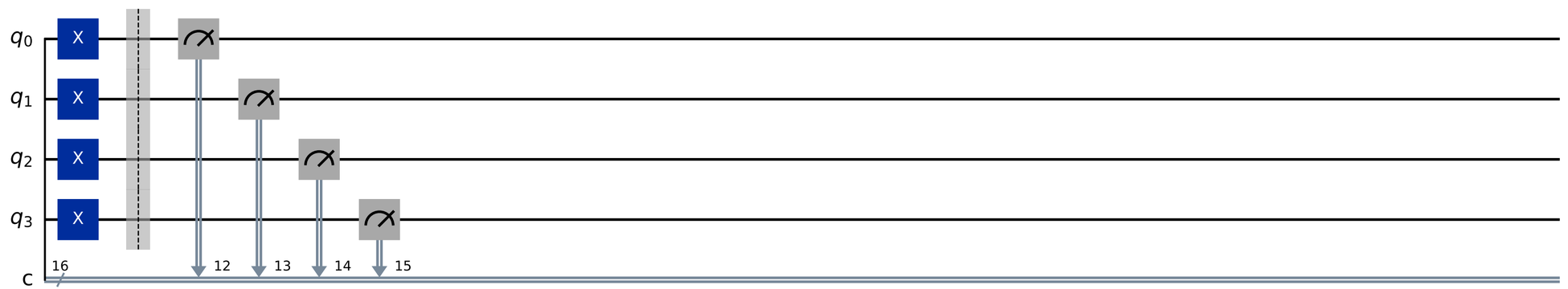

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00431
ZZ (0.00361)
ZZ (0.592)
R_{XX} 0.123
R_Y $-\pi/2$
|0⟩
3
2
1
0

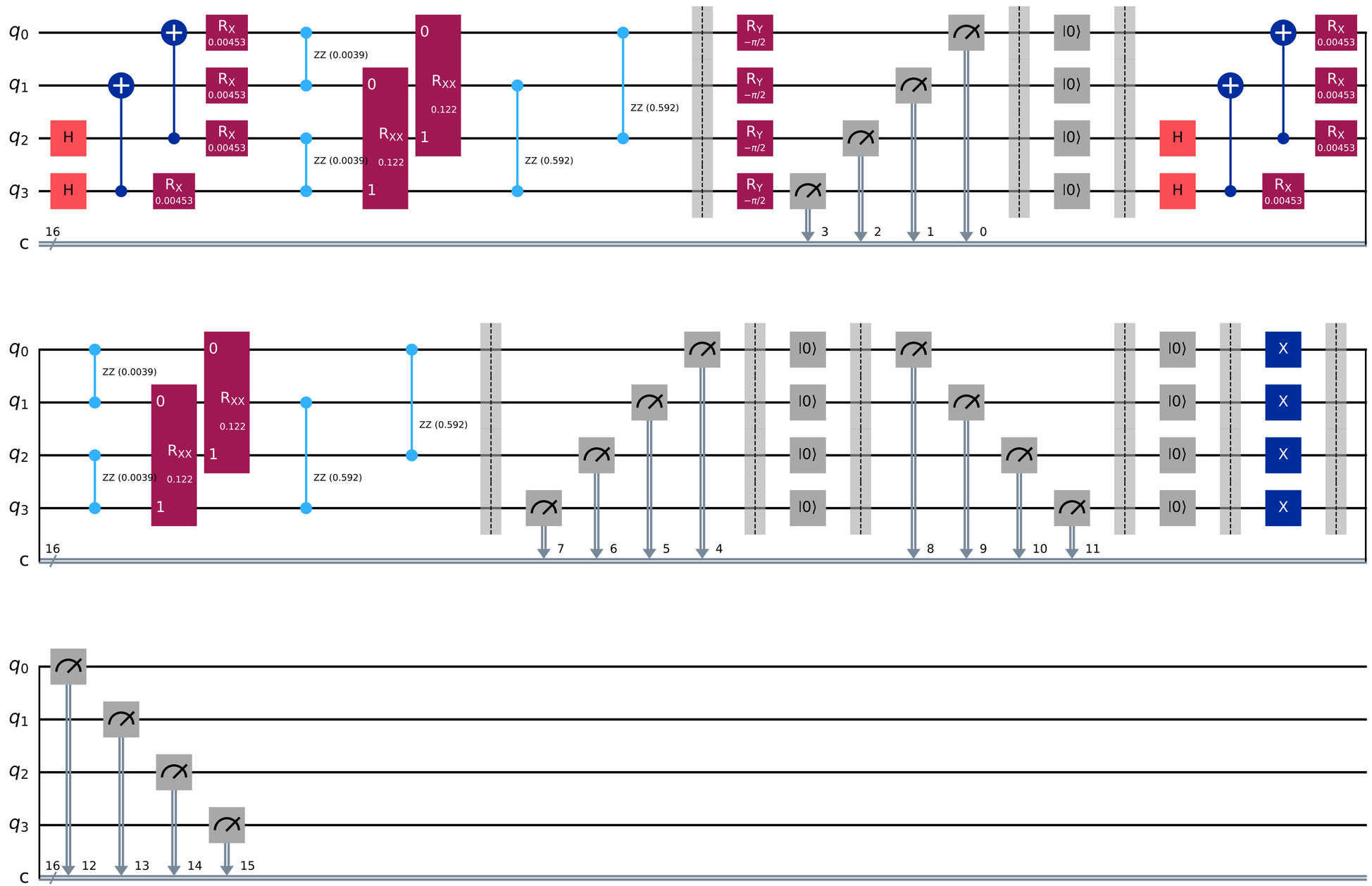

q_0
q_1
q_2
q_3
c
16
R_X 0.00431
ZZ (0.00361)
ZZ (0.592)
R_{XX} 0.123
|0⟩
7
6
5
4
8
9
10
11

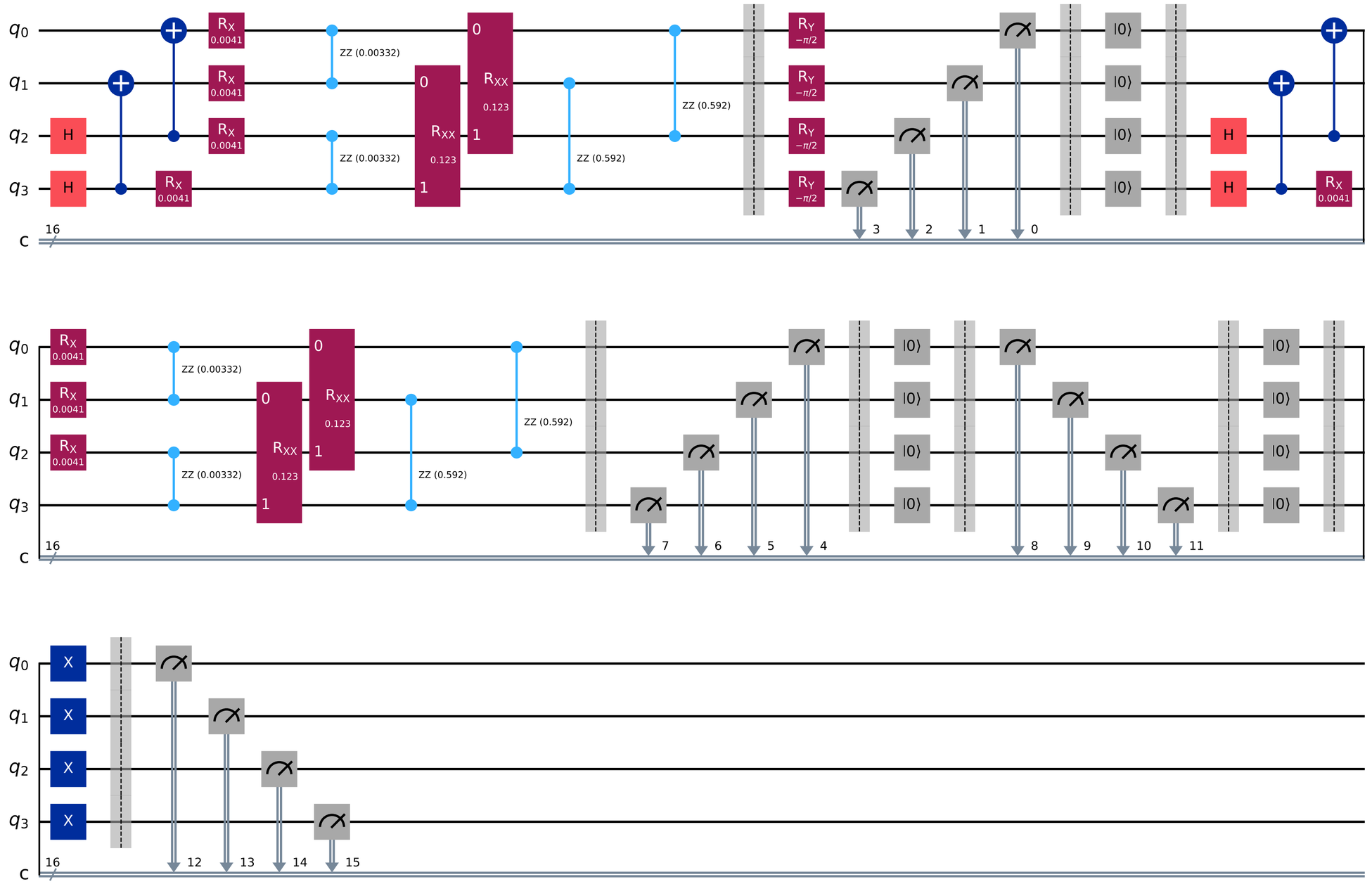

q_0
q_1
q_2
q_3
c
16
X
12
13
14
15

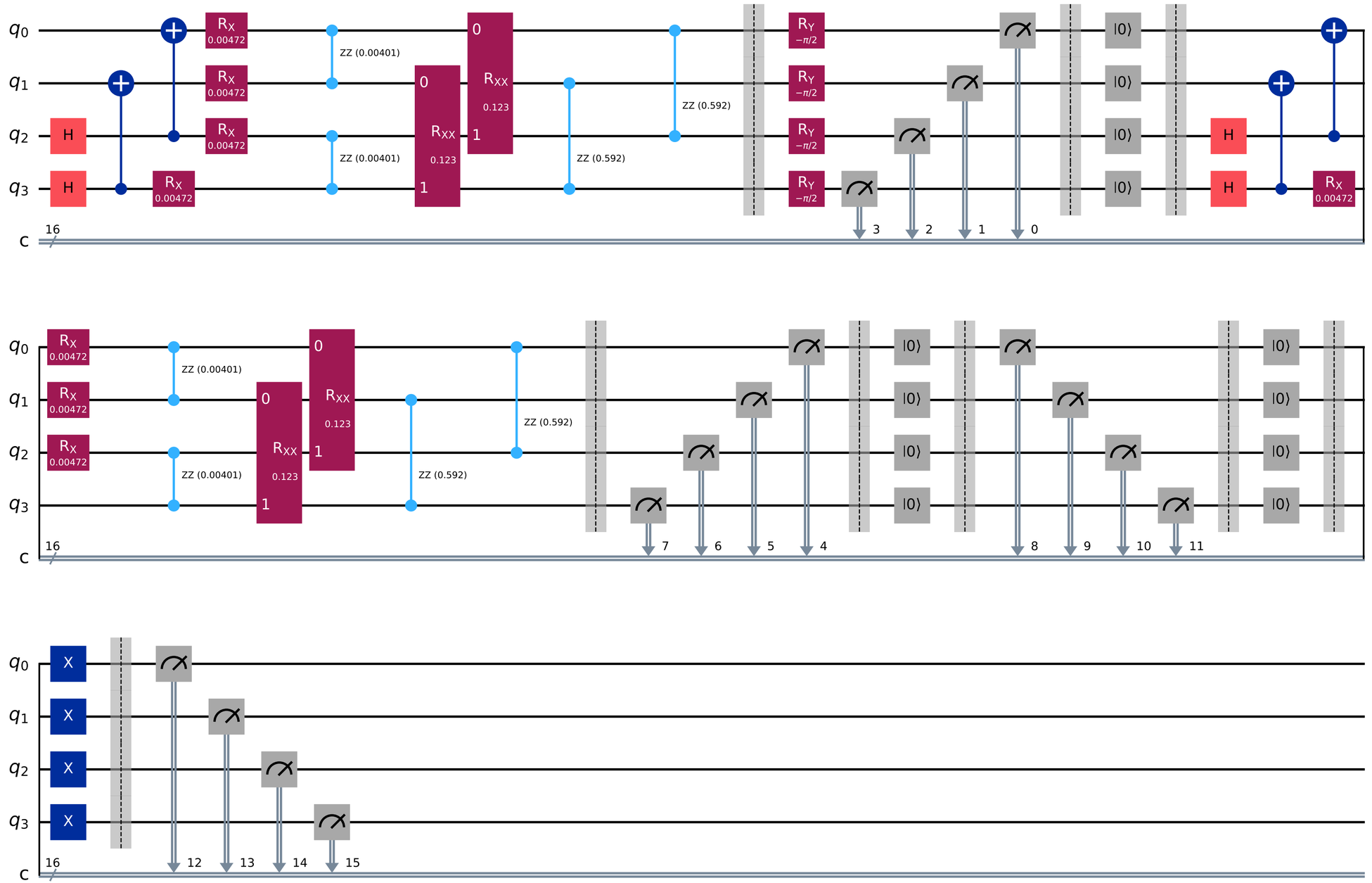

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00503
ZZ (0.00422)
ZZ (0.592)
R_{XX} 0.123
R_Y $-\pi/2$
|0⟩
3
2
1
0

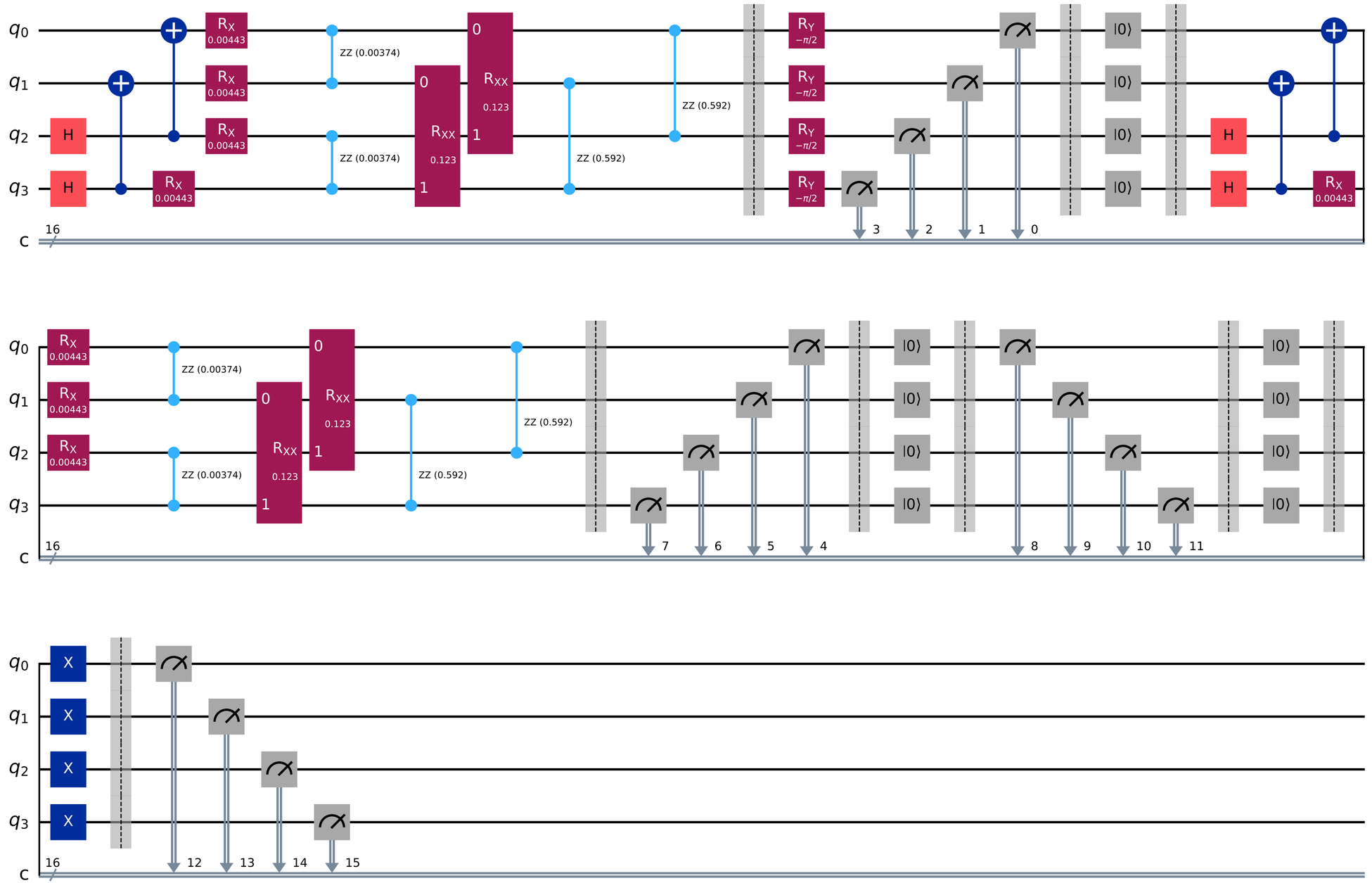

q_0
q_1
q_2
q_3
c
16
R_X 0.00503
ZZ (0.00422)
ZZ (0.592)
R_{XX} 0.123
|0⟩
7
6
5
4
8
9
10
11

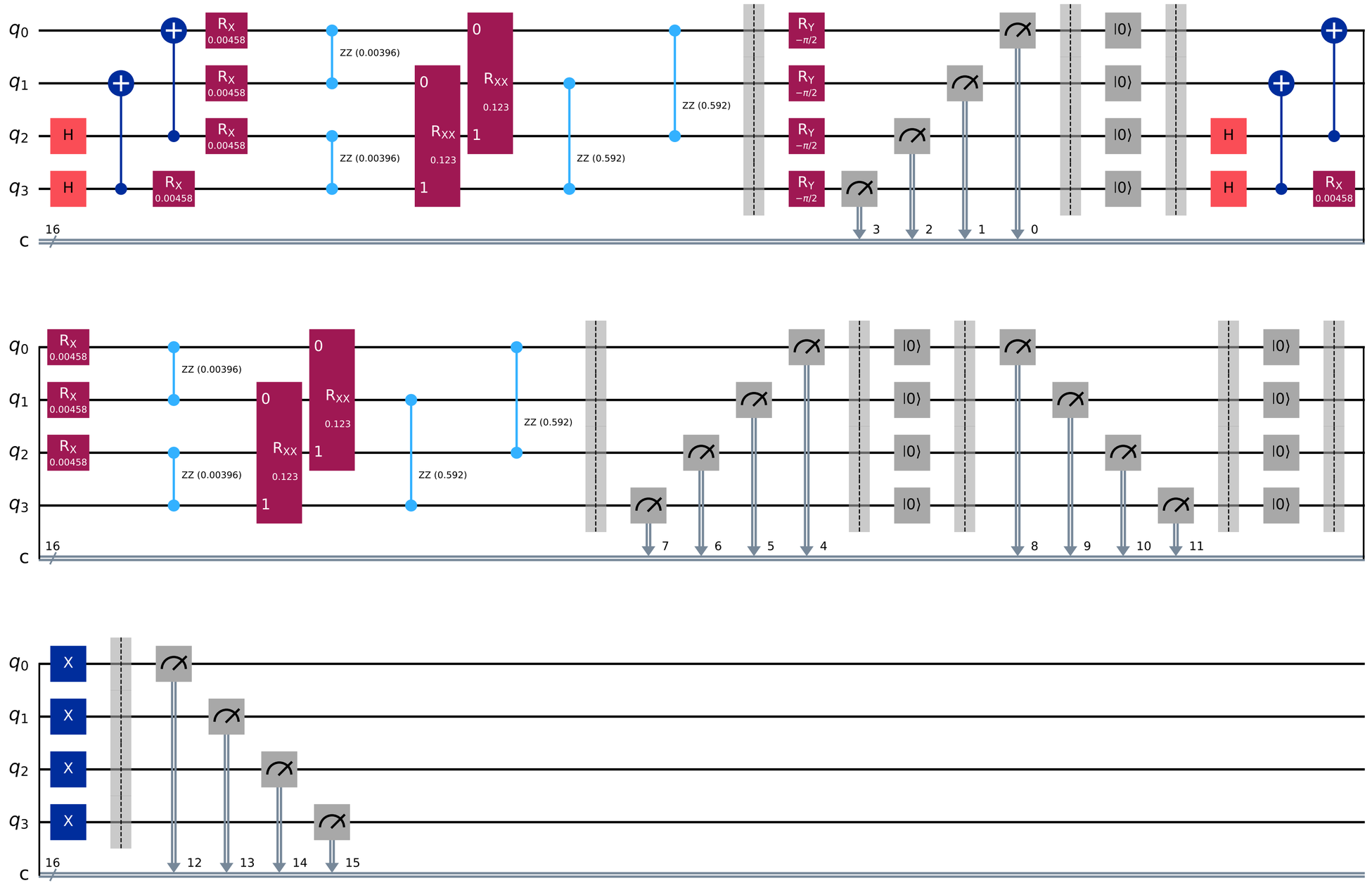

q_0
q_1
q_2
q_3
c
16
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX 0.00453
RX 0.00453
RX 0.00453
RX 0.00453
ZZ (0.0039)
ZZ (0.0039)
0
RXX 0.122
1
ZZ (0.592)
ZZ (0.592)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.00453
RX 0.00453
RX 0.00453
RX 0.00453
ZZ (0.0039)
ZZ (0.0039)
0
RXX 0.122
1
ZZ (0.592)
ZZ (0.592)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

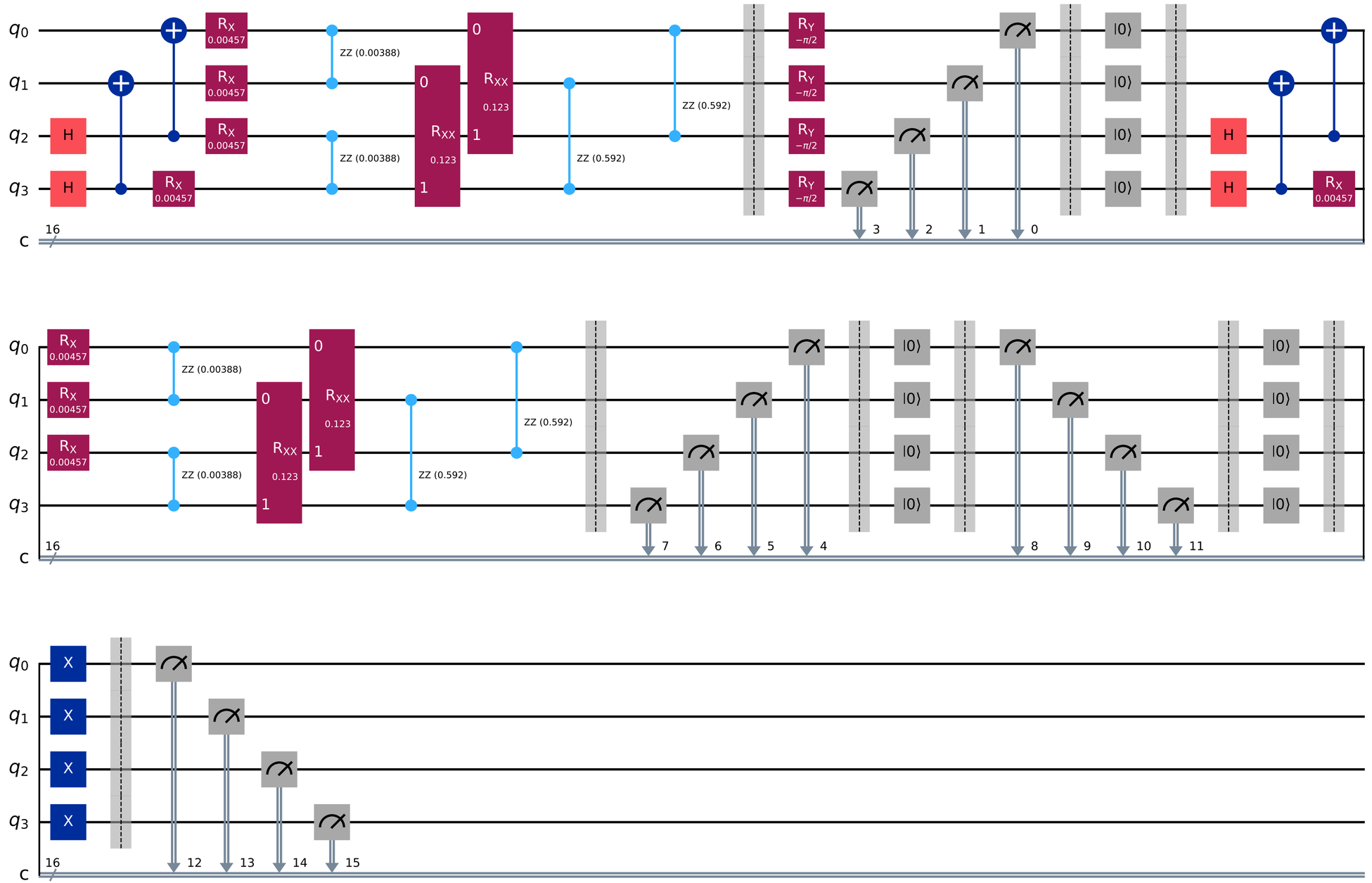

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0041
R_X 0.0041
R_X 0.0041
R_X 0.0041
ZZ (0.00332)
ZZ (0.00332)
0
R_{XX} 0.123
1
0
R_{XX} 0.123
1
ZZ (0.592)
ZZ (0.592)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.0041
3
2
1
0
R_X 0.0041
R_X 0.0041
R_X 0.0041
ZZ (0.00332)
ZZ (0.00332)
0
R_{XX} 0.123
1
0
R_{XX} 0.123
1
ZZ (0.592)
ZZ (0.592)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

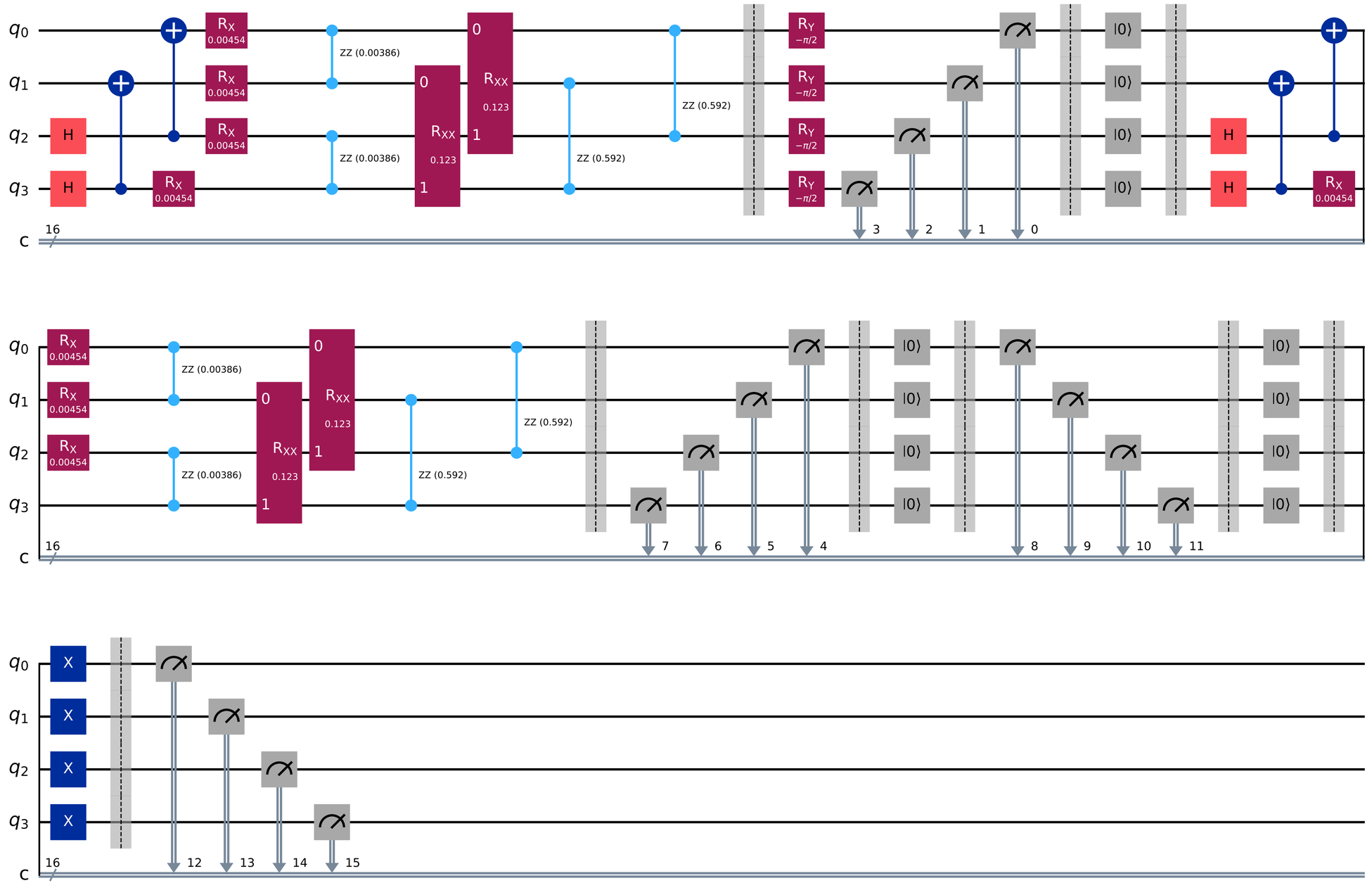

q0
q1
q2
q3
c
16
H
H
RX 0.00472
RX 0.00472
RX 0.00472
RX 0.00472
ZZ (0.00401)
ZZ (0.00401)
RXX 0.123
ZZ (0.592)
ZZ (0.592)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.00472
3
2
1
0
RX 0.00472
RX 0.00472
RX 0.00472
ZZ (0.00401)
ZZ (0.00401)
RXX 0.123
ZZ (0.592)
ZZ (0.592)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00443
R_Y $-\pi/2$
ZZ (0.00374)
ZZ (0.592)
R_{XX} 0.123
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00458
ZZ (0.00396)
ZZ (0.592)
R_{XX} 0.123
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
4
5
6
7
8
9
10
11
12
13
14
15

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00455
R_X 0.00455
R_X 0.00455
R_X 0.00455
ZZ (0.00387)
ZZ (0.00387)
0
R_{XX} 0.123
1
0
R_{XX} 0.123
1
ZZ (0.592)
ZZ (0.592)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.00455
3
2
1
0
R_X 0.00455
R_X 0.00455
R_X 0.00455
ZZ (0.00387)
ZZ (0.00387)
0
R_{XX} 0.123
1
0
R_{XX} 0.123
1
ZZ (0.592)
ZZ (0.592)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00457
ZZ (0.00388)
ZZ (0.592)
R_{XX} 0.123
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00454
R_X 0.00454
R_X 0.00454
R_X 0.00454
ZZ (0.00386)
ZZ (0.00386)
R_{XX} 0.123
ZZ (0.592)
ZZ (0.592)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.00454
3
2
1
0
R_X 0.00454
R_X 0.00454
R_X 0.00454
ZZ (0.00386)
ZZ (0.00386)
R_{XX} 0.123
ZZ (0.592)
ZZ (0.592)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

q0
q1
q2
q3
c
16
H
H
RX
0.00458
ZZ (0.00385)
ZZ (0.592)
RXX
0.123
RY
−π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X